

SEMANTICS FOR THE LOGIC OF ESSENCE

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ABSTRACT. This paper provides a possible worlds semantics for the system of the author's previous paper 'The Logic of Essence'. The basic idea behind the semantics is that a statement should be taken to be true in virtue of the nature of certain objects just in case it is true in any possible world compatible with the nature of those objects. It is shown that a slight variant of the original system is sound and complete under the proposed semantics.

KEY WORDS: essence, modality, possible worlds, semantics

In a previous paper 'The Logic of Essence', I presented a system for the logic of essence. In this paper, I present a semantics for a variant of the system and prove it complete with respect to that semantics.

The main primitive of the system is an indexed operator $\Box_F$. Where A is a sentence and F a predicate, $\Box_F A$ may be read: it is true in virtue of the nature of (some or all) the F 's that A . A statement of the form $\Box_F A$ is only taken to be true when each of the objects mentioned in A is involved in the nature of one of the F 's. Thus it is not taken to be true in virtue of the nature of the number 2 that Socrates = Socrates.

The system also contains symbols for what I call 'rigid predicates'. These are predicates that might be expressed in the form: x is either a_1 or a_2 or $\dots$, where $a_1, a_2, \dots$ are rigid names. Thus rigid predicates are essentially lists. These predicates are required in order to give a reasonable formulation of the reduction theses for the operator $\Box_F$.

The basic idea behind the semantics is that a statement should be taken to be true in virtue of the nature of certain objects if it is true in any world compatible with the nature of those objects. We shall make the simplifying assumption that each world is compatible with the nature of all and only those objects that it contains. Thus the condition for a statement to be true in virtue of the nature of certain objects is that it should be true in all those worlds that contain those objects. However, the presence of an object in a world is not taken to guarantee its existence but merely its possibility. Thus each world will be taken to embody its own 'view' of which objects are possible and which are not.

The proof of completeness for the system is not straightforward. The difficulties essentially arise from the presence of rigid predicates. For in



constructing a canonical model, we will need to add new ‘witnessing’ predicates. But since we cannot quantify over these predicates, it is hard to maintain control on the consistency of the resulting construction. One novelty of the proof is a technique for eliminating the predicates through approximating definitions rather than through quantification. Another is a device for getting the canonical model into ‘equilibrium’ by splitting it into parts.

The first two sections are devoted to the language of the logic (Section 1) and the system of proof (Section 2). The next section gives the semantics (Section 3). The remaining six sections develop the completeness proof. The first three (Sections 4–6) provide lemmas crucial to the construction of the canonical model. The next two sections (Sections 7, 8) show how to build up a ‘diagram’ of the model; and the last section (Section 9) shows how to obtain the model itself. The reader might find it helpful to have the previous paper ‘The Logic of Essence’ at hand (henceforth abbreviated to ‘LE’) and also to consult the papers ‘Essence and Modality’ and ‘Senses of Essence’ for further explanation of the notion of essence and for general philosophical motivation.¹

1. REVIEW OF THE LANGUAGE

The vocabulary of the system consists of the following symbols:

- (i) denumerably many individual variables;
- (ii) denumerably many individual constants;
- (iii) denumerably many n -place pure predicate symbols, for $n = 0, 1, 2, \dots$;
- (iv) denumerably many 1-place rigid predicate symbols;
- (v) the 2-place identity predicate $=$, and the 2-place dependency predicate $\geq$;
- (vi) the logical constants $\sim$, $\vee$, and $\prod$;
- (vii) the essentialist operator symbol $\Box$;
- (viii) the abstraction operator λ ;
- (ix) the bracketing devices $(,)$, $[$ and $]$, and $\therefore$.

The language of LE did not contain constants, but they are useful for the semantics and for the proof of completeness. The ‘pure’ predicates are ones involving no reference to any object. We use ‘r-predicate symbol’ as shorthand for ‘rigid predicate symbol’.

The *formulas* and *predicates* of the system are defined by the following simultaneous induction:

- (i) if F is an n -place predicate and $t_1, \dots, t_n$ are variables or constants, then $Ft_1 \dots t_n$ is a formula;
- (ii) standard rules for $\sim$ and $\vee$ and for the quantifier $\prod$.
- (iii) If A is a formula and F is a 1-place predicate, then $\Box[F : A]$ is a formula;
- (iv) Each n -place predicate symbol is an n -place predicate;
- (v) If A is a formula and x a variable, then $\lambda x A$ is a 1-place predicate.

We adopt standard meta-linguistic terminology and conventions. In particular, we use:

- (i) $\top$ for $\prod x(x = x)$, where x is the first free variable under some standard ordering of the variables;
- (ii) $\perp$ for $\sim \top$;
- (iii) $\bigvee_{1 \leq i \leq n} A_i$ for $\perp$ if $n = 0$, for A_1 if $n = 1$, and for $(\dots (A_1 \vee A_2) \vee \dots \vee A_n)$ if $n > 1$ (and similarly for the generalized conjunction $\bigwedge_{1 \leq i \leq n} A_i$).

We use F and G etc. for arbitrary 1-place predicates, P, Q, R and S etc. for arbitrary r -predicate symbols, and H for an arbitrary n -place predicate, $n \geq 0$. We use boldface to indicate a list of symbols. Thus $\mathbf{x}$ indicates a list of n variables $x_1, \dots, x_n$, $n \geq 0$; and similarly for $\mathbf{F}$ and $\mathbf{P}$. In the formula $\Box[F : B]$, the predicate F is called the *delimiter*.

A predicate is said to be *rigid* if it is either an r -predicate symbol or is of the form $\lambda x \bigvee_{1 \leq i \leq n} A_i$, $n \geq 0$, where each formula A_i , $i = 1, \dots, n$, is either of the form Px or of the form $x = y$ for some variable y distinct from x .

We adopt the following further abbreviations:

- (I) *Variants on* $\Box$.
 - (i) $\Box_F A$ for $\Box[F : A]$;
 - (ii) $\Diamond[F : A]$ (and $\Diamond_F A$) for $\sim \Box[F : \sim A]$.
- (II) *Dependency Notions*
 - (i) $x \leq y$ for $y \geq x$;
 - (ii) $x \leq F$ for $\Sigma y (Fy \ \& \ x \leq y)$, y the first variable distinct from x and not free in F ;
 - (iii) $\text{Cl}(F)$ for $\prod x \prod y (Fx \ \& \ x \geq y \rightarrow Fy)$.

$\text{Cl}(F)$ states that F is closed under the relation of dependence.

(III) *Constituency*

Let E be a formula or predicate. Suppose that $x_1, \dots, x_m$ are the free variables of E in order of appearance and that $P_1, \dots, P_n$ are the r -predicate symbols of E , likewise in order of appearance. We then use:

- (i) $x\eta E$ for $\bigvee_{1 \leq i \leq m} x = x_i \vee \bigvee_{1 \leq j \leq n} P_j x$.

The intuitive meaning of $Hx\eta E$ is that x occurs in the proposition (or relation) expressed by E .

(IV) *Predicational Notions*

- (i) $\bigvee$ for $\lambda x \top$;
- (ii) $\bigwedge$ for $\lambda x \perp$;
- (iii) (y) for $\lambda x (x = y)$, x the first variable distinct from y ;
- (iv) $|E|$ for $\lambda x (x\eta E)$, x the first variable not free in E ;
- (v) cF for $\lambda x (x \leq F)$, x the first variable not free in F ;
- (vi) $(F_1, \dots, F_n)$ for $\lambda x \bigvee_{1 \leq i \leq n} F_i x$, where $n \geq 0$ and x is the first variable not to occur free in any of $F_1, \dots, F_n$;
- (vii) $F - G$ for $\lambda x (Fx \& \sim Gx)$;
- (viii) $F \subset G$ for $\prod x \sim (Fx \& \sim Gx)$, where F and G are 1-place predicates and x is the first variable not to occur free in F or G ;
- (ix) $R \approx S$ for $(R \subset S \& S \subset R)$.

The reader should distinguish between the object-language $\subset$ and the meta-language $\subseteq$. In using (t), t a variable or constant, as part of a delimiter we will usually omit the brackets. Outermost brackets of the disjunctive predicate $(F_1, \dots, F_n)$ under (vi) may also be omitted. Thus, combining these two conventions, we may use $\Box_{x,y} A$ to abbreviate $\Box_{\lambda z((x)z \vee (y)z)} A$.

2. REVIEW OF THE SYSTEM

The axioms and rules of the system E5 are as follows:

(I) *Standard Classical Axioms and Rules:*

We may use any of the standard axiomatizations of classical first-order logic. However, we shall suppose that the sole rules of inference are Modus Ponens (MP) and Universal Generalization (Gen).

(II) *Modal Axioms and Rules:*

- (i) $\Box_F A \rightarrow A$;
- (ii) $\Box_F (A \rightarrow B) \rightarrow (\Box_F A \rightarrow \Box_F B)$;
- (iii) $\sim \Box_F A \rightarrow \Box_{F,|A|} \sim \Box_F A$, F rigid;
- (iv) $A \setminus \Box_{|A|} A$;
- (v) $F \subset G \rightarrow (\Box_F A \rightarrow \Box_G A)$;

(III) *Predicational Axioms and Rules:*

- (i) $\lambda x A y \leftrightarrow A(y)$, where $A(y)$ is the result of substituting y freely for free occurrences of x in A ;

- (ii) $Px \rightarrow \Box_P Px$;
- (iii) (Predicate Elimination) $P \approx F \rightarrow A/A$, as long as the predicate P does not occur in A or F .
- (iv) (Modal Predicate Elimination) $A \rightarrow (Q \subset P \rightarrow \Box_P \prod \mathbf{x}(Q \approx F \rightarrow B))/A \rightarrow \Box_P \prod \mathbf{x}(F \subset P \rightarrow B)$, where Q is distinct from P and does not occur in F or B .

(IV) *Dependency Axioms*:

- (i) $\Box_{cF} A \rightarrow \Box_F A$;
- (ii) $\Box_F A \& x\eta A \rightarrow x \leq F$;

Rule (III)(iv) was not included in the original formulation of E5. It seems necessary for the proof of completeness.

Suppose we add quantifiers for r -predicate symbols to the language and then add standard axioms and rules for these quantifiers to the system, plus the axiom:

ΣP : $\Sigma P(P \approx F)$, for P not free in F .

The rules (III)(iii) and (iv) can then be derived. For suppose $\vdash P \approx F \rightarrow A$. By quantificational reasoning, $\vdash \Sigma P(P \approx F) \rightarrow A$ and so, by ΣP and Modus Ponens, $\vdash A$. Now suppose $\vdash A \rightarrow (Q \subset P \rightarrow \Box_P \prod \mathbf{x}(Q \approx F \rightarrow B))$. Then, by quantificational reasoning, $\vdash A \rightarrow \prod Q(Q \subset P \rightarrow \Box_P \prod \mathbf{x}(Q \approx F \rightarrow B))$. But then by modal reasoning (as in Theorem V.1 of LE), $\vdash A \rightarrow \Box_P \prod Q(Q \subset P \rightarrow \prod \mathbf{x}(Q \approx F \rightarrow B))$, hence $\vdash A \rightarrow \Box_P \prod \mathbf{x} \prod Q(Q \subset P \rightarrow (Q \approx F \rightarrow B))$, and hence: (a) $\vdash A \rightarrow \Box_P \prod \mathbf{x}(\Sigma Q(Q \subset P \& Q \approx F) \rightarrow B)$. From axiom ΣP , $\vdash \Box_P \prod \mathbf{x} \Sigma Q(Q \approx F)$, and hence: (b) $\vdash \Box_P \prod \mathbf{x}(F \subset P \rightarrow \Sigma Q(Q \subset P \& Q \approx F))$. But then from (a) and (b), $\vdash A \rightarrow \Box_P \prod \mathbf{x}(F \subset P \rightarrow B)$.

The rules (iii) and (iv) appear to be the price we must pay in order to avoid quantification over r -predicates. However, I do not know whether either or both of the rules could be dropped without loss of theorems.

3. SEMANTICS FOR THE SYSTEM

A *model* M is a quadruple $\langle W, I, \succeq, \phi \rangle$, where:

- (i) W (worlds) is a non-empty set;
- (ii) I (individuals) is a function taking each $w \in W$ into a non-empty set I_w ;
- (iii) $\succeq$ (dependence) is a reflexive and transitive relation on $\bigcup_{w \in W} I_w$ with respect to which each world is closed ($a \in I_w$ and $a \succeq b$ implies $b \in I_w$);

- (iv) ϕ (valuation) is a function that takes each constant a into an individual $\phi(a)$ of some I_w ($w \in W$), each r -predicate symbol H into a subset $\phi(H)$ of some I_w , and each world w and pure n -place predicate symbol F into a set $\phi(F, w)$ of n -tuples of I_w .

Given any subset J of $\bigcup I_w$, we let the *closure* $c(J)$ of J (in M) be $\{b : a \succeq b \text{ for some } a \in J\}$. Thus the subset J is closed if $c(J) \subseteq J$. It is readily verified that c satisfies the standard closure conditions: $J \subseteq c(J)$; $cc(J) \subseteq c(J)$; and $c(J_1) \cup c(J_2) = c(J_1 \cup J_2)$.

Let M be a model and E a sentence or closed predicate whose constants are $a_1, \dots, a_m$ and whose r -predicate symbols are $P_1, \dots, P_n$. Then the *objectual content* $[E]^M$ of E (in M) is taken to be $\{\phi(a_1), \dots, \phi(a_m)\} \cup \phi(P_1) \cup \dots \cup \phi(P_n)$; and we say that E is *defined* (in M) at $w \in W$ if $[E]^M \subseteq I_w$. The index M will usually be dropped from $[E]^M$ and from other model-indexed terms.

The model M is said to be *full* if for each $a \in \bigcup I_w$ there is a constant a for which $\phi(a) = a$. Given a full model M , we now define $w \models A$, the *truth* of the sentence A at w , and H_w , the *extension* of the predicate H at w . We shall presuppose that $w \models A$ is informally defined only when A is (in the technical sense) defined in M at w and that H_w is defined only when H is defined at w . The clauses are then as follows:

- (i) (a) $w \models Ha_1 \dots a_n$ iff $\langle \phi(a_1), \dots, \phi(a_n) \rangle \in H_w$, where H is any non-logical predicate;
- (i) (b) $w \models a = b$ iff $\phi(a) = \phi(b)$;
- (i) (c) $w \models a \succeq b$ iff $\phi(a) \succeq \phi(b)$;
- (ii) $w \models \sim A$ iff not $w \models A$;
- (iii) $w \models (A \vee B)$ iff $w \models A$ or $w \models B$;
- (iv) $w \models \prod x A(x)$ iff $w \models A(a)$ whenever $\phi(a) \in I_w$;
- (v) $w \models \Box_F A$ iff (a) $[A]^M \subseteq c(F_w)$ and (b) $v \models A$ whenever $I_v \supseteq F_w$;
- (vi) $F_w = \phi(w, F)$;
- (vii) $P_w = \phi(P)$;
- (viii) $\lambda x A(x)_w = \{a \in I_w : w \models A(a)\}$.

Truth and extension for non-full models may be defined by extending the language and the model with a new constant for each individual of the model. We omit the obvious details.

We cite, without proof, the following elementary results, whose use will often be tacit:

LEMMA 1. *Given any model M and world w in M ,*

- (i) $[\sim A] = [A]$, $[(A \vee B)] = [A] \cup [B]$, $[\Box_F A] = [F] \cup [A]$, and $[|A|] = [A]$;

- (ii) $[E] = |E|_w$ for any sentence or closed predicate E and any world w at which E is defined;
- (iii) for F a closed rigid predicate, $[F] = F_w$ whenever F is defined at w ;
- (iv) $(cF)_w = c(F_w)$ whenever F is defined at w .

A sentence A is said to be *valid* if for every model M and every world w of M at which A is defined, A is true at w in M ; and a formula $A(x_1, \dots, x_n)$ is said to be *valid* if each of its closed instance $A(a_1, \dots, a_n)$ is valid. The notions of *consequence* and of *satisfiability* are defined similarly.

THEOREM 2 (Soundness). *Each theorem of E5 is valid.*

Proof. We prove by induction that each closed instance of each theorem is valid:

(I) *Standard Classical Axioms and Rules.* Straightforward (though the case of Modus Ponens needs to be treated with some care).

(II) *Modal Axioms and Rules.*

(i) Suppose $w \models \Box_F A$. By clause (v)(b) in the truth-definition, $v \models A$ whenever $I_v \supseteq F_w$. But clearly $I_w \supseteq F_w$; and so $w \models A$.

(ii) Suppose $w \models \Box_F(A \rightarrow B)$ and $w \models \Box_F A$. Then $[(A \rightarrow B)] \subseteq c(F_w)$, and both $v \models (A \rightarrow B)$ and $v \models A$ whenever $I_v \supseteq F_w$. But then $[B] \subseteq c(F_w)$ by Lemma 1(i) and $v \models B$ whenever $I_v \supseteq F_w$; and so $w \models \Box_F B$.

(iii) Suppose $w \models \sim \Box_F A$, for F rigid. By Lemmas 1(i) and (iii), $[A] \subseteq |A|_w$ and $[F] \subseteq F_w$; and so $[\sim \Box_F A] \subseteq c((F, |A|)_w)$. We now distinguish two cases. (a) $[A] \not\subseteq c(F_w)$. Take any v for which $I_v \supseteq F_w$. Then F is defined at v and so, by Lemma 1(iii), $[A] \not\subseteq c(F_v)$. But then $v \models \sim \Box_F A$ for any such v ; and so $w \models \Box_{F, |A|} \sim \Box_F A$. (b) $[A] \subseteq c(F_w)$. Then $v \models \sim A$ for some v for which $I_v \supseteq F_w$. Take now any u for which $I_u \supseteq F_w \cup |A|_w$. Then F is defined at u , with $F_u = F_w$, and so $I_v \supseteq F_u$. So given $v \models \sim A$, $u \models \sim \Box_F A$ and hence $w \models \Box_{F, |A|} \sim \Box_F A$.

(iv) Suppose A is valid. Consider any model M and any world w of M at which A is defined. Then $[A] \subseteq |A|_w$ and, for any v for which $I_v \supseteq |A|_w = [A]$, A is defined and so $v \models A$. But then $w \models \Box_{|A|} A$ for any such w and so $\Box_{|A|} A$ is valid.

(v) Suppose $w \models F \subset G$ and $w \models \Box_F A$. Then $[A] \subseteq c(F_w)$ and $v \models A$ whenever $I_v \supseteq F_w$. Since $F_w \subseteq G_w$, $c(F_w) \subseteq c(G_w)$; and so $[A] \subseteq c(G_w)$. Take now any v for which $I_v \supseteq G_w$. Then since $G_w \supseteq F_w$, $I_v \supseteq F_w$ and $v \models A$. Therefore $w \models \Box_G A$.

(III) *Predicational Axioms and Rules.*

(i) Straightforward.

(ii) Suppose $w \models Pa$. Then $\phi(a) \in \phi(P)$; and so $[Pa] \subseteq P_w = \phi(P) \subseteq$

$c(P_w)$. Taken any v for which $I_v \supseteq P_w$. Then clearly $v \models Pa$; and so $w \models \Box_P Pa$.

(iii) Suppose A is not valid. Then for some model M and w in M at which A is defined, not $w \models A$. The domain I_w contains an individual a . Let M' be the model like M except that each constant in F but not in A refers to a , each r -predicate symbol P in F but not in A has the null set as its extension, and P has the extension of F at w as its extension. Then $P \approx F \rightarrow A$ is defined at w in M' , $w \models P \approx F$ in M' and not $w \models A$ in M' . Therefore $P \approx F \rightarrow A$ is not valid.

(iv) Suppose the conclusion of the rule is not valid. Then for some model M and w in M , $w \models A$ and $w \models \sim \Box_P [\Box \mathbf{x}(F(\mathbf{x}) \subset P \rightarrow B(\mathbf{x}))]$ (where the variables $\mathbf{x}$ in F and B are now displayed). We distinguish two cases. (a) $[\Box \mathbf{x}(F(\mathbf{x}) \subset P \rightarrow B(\mathbf{x}))] \not\subseteq cP_w$. Then $[\Box \mathbf{x}(P \approx F(\mathbf{x}) \rightarrow B(\mathbf{x}))] \not\subseteq cP_w$; and so $w \models \sim(A \rightarrow (P \subset P \rightarrow \Box_P [\Box \mathbf{x}(P \approx F(\mathbf{x}) \rightarrow B(\mathbf{x}))]))$, and $A \rightarrow (Q \subset P \rightarrow \Box_P [\Box \mathbf{x}(Q \approx F(\mathbf{x}) \rightarrow B(\mathbf{x}))])$ is not valid. (b) $[\Box \mathbf{x}(F(\mathbf{x}) \subset P \rightarrow B(\mathbf{x}))] \subseteq cP_w$. Then $v \models \sim [\Box \mathbf{x}(F(\mathbf{x}) \subset P \rightarrow B(\mathbf{x}))]$ for some v with $I_v \supseteq P_w$; and so there are $a_1, \dots, a_n \in I_v$ with respective names $a_1, \dots, a_n$ for which $v \models F(a_1, \dots, a_n) \subset P$ and $v \models \sim B(a_1, \dots, a_n)$. Let J be $F(a_1, \dots, a_n)_v$ and let M' be the model whose only difference from M is that J is assigned as the extension of Q . Since $v \models F(a_1, \dots, a_n) \subset P$ in M , $w \models Q \subset P$ in M' . By stipulation, $v \models Q \approx F(a_1, \dots, a_n)$ in M' ; and, since $v \models \sim B(a_1, \dots, a_n)$ in M , $v \models \sim B(a_1, \dots, a_n)$ in M' . But then $w \models \sim \Box_P [\Box \mathbf{x}(Q \approx F(\mathbf{x}) \rightarrow B(\mathbf{x}))]$ in M' ; and so $A \rightarrow (Q \subset P \rightarrow \Box_P [\Box \mathbf{x}(Q \approx F(\mathbf{x}) \rightarrow B(\mathbf{x}))])$ is not valid.

(IV) Dependency Axioms.

(i) Suppose $w \models \Box_{cF} A$. Then $[A] \subseteq c((cF)_w) = cc(F_w) = c(F_w)$ by Lemma 1(iv). Take now any v for which $I_v \supseteq F_w$. Then from the closure condition and Lemma 1(iv) again, $I_v = c(I_v) \supseteq c(F_w) = (cF)_w$. So $v \models A$; and hence $w \models \Box_F A$.

(ii) Suppose $w \models \Box_F A \& b \eta A$. Then $[A] \subseteq (cF)_w = c(F_w)$ and $\phi(b) \in [A]$. But then, for some $a \in F_w$, $a \geq b$; and so $w \models b \leq F$.

It is interesting to note that there is an interpretation of the formal semantics in which each domain I_w is taken to consist of the objects that *exist* at w . A modal statement $\Box_F A$ is then true at a world w iff each object mentioned in A depends for its existence on something that F 's and A is true in each world that contains the objects that exist in F . The proof to follow will then establish that the system is complete under this modal-existential interpretation of $\Box_F$. However, it would be a serious mistake to suppose that the notion of essence can therefore be analyzed in terms of modality and existence. Two notions can have the same logic, even the same formal semantics, and yet be intuitively quite different. Indeed, under

the proposed interpretation, each existent will essentially exist and yet this is not a result which, intuitively, we want. Or again, it is hard to see under what sense of necessity the existence of $\{2\}$ would entail that $2 \in \{2\}$ and yet the existence of 2 not entail that $2 \in \{2\}$. $\square$

4. SYNTACTIC PRELIMINARIES

We present some of the basic notions required for the completeness proof and the motivation for using them.

Languages. It will be important to keep track of the languages of the different worlds in the canonical model. We take a *language* to be a set of constants and r-predicate symbols. The *language* $\mathcal{L}(A)$ of a formula A is the set of individual constants and r-predicate symbols that occur in A . Similarly for the *language of* a predicate or *of* a set of formulas. We shall sometimes loosely talk of a sentence or predicate being in a given language $\mathcal{L}$ when we mean that its own language is included in the language $\mathcal{L}$.

Original Predicates. We shall suppose that a certain subclass of *original* individual constants and predicate symbols is specified in advance. The given predicate symbols are to include *all* of the pure predicate symbols and all but infinitely many r-predicate symbols; and, likewise, the given constants are to include all but infinitely many constants of the language. It is from this subclass of symbols that we draw the consistent set of sentences that is to be proved satisfiable. The remaining constants and r-predicate symbols are placed on reserve, to be used as ‘witnesses’ in the completeness proof. A predicate symbol is said to be *original* if it is from the given subclass and to be *supplemental* or an *s-predicate* symbol otherwise. A formula, predicate, or set of formulas is said to be *original* if all of its predicate symbols are original and is said to be *supplemental* otherwise. Note that no restriction is placed on which individual constants can occur in an the original formula.

Restricted Formulas. We shall impose a global restriction on which supplemental sentences can be used in the worlds of the canonical model. The presence of sentences satisfying this restriction is required to establish that the canonical model has the required properties.

A *basic restricted* sentence is one of the form:

- (i) $F \subset S$;
- (ii) $S \subset F$;
- (iii) Sa ,

where S is an s-predicate symbol and F is either an s-predicate symbol or an original closed predicate.

A *restricted* sentence is then either an original sentence, a basic restricted sentence, a truth-functional compound of basic restricted sentences and original sentences, or a sentence of the form:

- (iv) $\Box_S A$, for S an s-predicate symbol and A an original sentence.

We say that a restricted sentence B is *negatable* if $\sim B$ is also a restricted sentence and that otherwise it is *unnegatable*. It should be evident that the unnegatable restricted sentences are exactly those sentences of the form $\Box_S A$ listed under (iv). A set of sentences is said to be *restricted* if each of its members is restricted. From henceforth we shall use A (and its variants) for arbitrary unrestricted formulas, B , C , and D for restricted formulas, and Δ and Γ for sets of restricted sentences unless there is some explicit indication to the contrary.

Completeness Concepts. A set of sentences Δ is said to be:

- (i) *consistent* if not $\Delta \vdash \perp$;
- (ii) a *theory* if $B \in \Delta$ whenever $\Delta \vdash B$ and B is in $\mathcal{L}(\Delta)$;
- (iii) *N-complete* if $B \in \Delta$ or $\sim B \in \Delta$ for any negatable B in $\mathcal{L}(\Delta)$;
- (iv) *Q-complete* if $B(a) \in \Delta$ for some individual constant whenever a sentence $\Sigma x B(x) \in \Delta$;
- (v) *standardly complete* if both N-complete and Q-complete;
- (vi) *P-complete* if $Q \approx F$ for some s-predicate symbol Q whenever F is an original closed predicate of $\mathcal{L}(\Delta)$;
- (vii) *fully complete* if standardly complete and P-complete.

Note the restriction in definitions (ii) and (iii) to the language of the given set Δ . Thus if $\prod x B(x)$ belongs to the theory Δ then $B(a)$ will not belong to Δ unless a is in the language of Δ ; and an N-complete theory need only ‘decide’ a sentence when it is negatable and in the language of the theory.

A set of sentences Δ is said to *extend* or be an *extension* of another Δ' if $\Delta \supseteq \Delta'$ and it is a *safe* extension of Δ if $\mathcal{L}(\Delta)$ contains no original r-predicates not already in $\mathcal{L}(\Delta')$. Δ' is said to *subsume* Δ if $\Delta' \vdash A$ for every sentence A in Δ ; and Δ and Δ' are (*deductively*) *equivalent* if each subsumes the other.

Companionship. Worlds accessible from one another in the canonical model will be called companions. To be more precise, if P is an s-predicate symbol, then the set of sentences Γ is said to be a *P-companion* to the set of sentences Δ if $P \in \mathcal{L}(\Delta)$ and if, for any arbitrary sentence A in $\mathcal{L}(\Delta)$ for

which $\Delta \vdash \Box_p A$, $\Gamma \cup \{A\}$ is consistent. Note that the sentence A in this definition need not be restricted. If Γ is a P-companion to Δ , then both Γ and Δ must be consistent. For the sentence $\top$ is in $\mathcal{L}(\Delta)$ and $\Delta \vdash \Box_p \top$. So, if Γ is a P-companion to Δ , $\Gamma \cup \{\top\}$ is consistent and so Γ itself is consistent. On the other hand, if Δ is inconsistent, then $\perp$ is in $\mathcal{L}(\Delta)$ and $\Delta \vdash \Box_p \perp$. But $\Gamma \cup \{\perp\}$ is inconsistent and so Γ is not a P-companion to Δ .

Γ is said to be a *maximal* P-companion to Δ if it is a P-companion and if no proper extension of Γ in the language of Γ is a P-companion to Δ ; and Γ is a *broad* P-companion to Δ if it is a P-companion and if $B \in \Gamma$ whenever $\Gamma \cup \{A : \Delta \vdash \Box_p A \text{ and } A \text{ is in } \mathcal{L}(\Delta)\} \vdash B$ and B is in $\mathcal{L}(\Gamma)$. It should be clear that every maximal P-companion is broad and that every broad P-companion is a theory.

It is essential in the definition of companionship that the sentence A be required to belong to the language $\mathcal{L}(\Delta)$. For we should allow $\{\Sigma x \sim Px, \Diamond \Box_p \prod x Px\}$ to be consistent. But this will require (mutual) P-companions Δ and Γ with $\sim Pa, \Diamond \Box_p \prod x Px \in \Delta$ and $\Box_p \prod x Px \in \Gamma$. Thus even though $\prod x Px$ is inconsistent with $\sim Pa$, we still wish Δ to be a P-companion to Γ .

Definitions. These are the means by which predicates in the canonical model are witnessed. Witnesses, in the form of rigid predicates, are required in order to establish that the canonical model has the appropriate S5-structure. Formally speaking, a *definition* is a sentence of the form $Q \approx F$, for Q an s-predicate symbol and F a closed predicate. Q is said to be its *definiendum* and F its *definiens*, and a definition is said to be *nonmodal* when its definiens is nonmodal. Unless otherwise stated, it will be assumed that F is an original predicate (containing no s-predicate symbols). A *system* of definitions is a set of definitions no two of which share a common definiendum. A system of definitions is said to be of its *definiendia* and to be *in terms* of the predicate symbols that occur in its *definiendia*. We say that an s-predicate symbol Q is *defined* in the set of sentences Δ if $Q \approx F \in \Delta$ for some original predicate F and that otherwise it is *undefined*.

5. DEFINITIONAL REPLACEMENT

We show how a set of restricted sentences may be approximated by means of a system of definitions. We then consider two significant ways in which this result is of use in establishing the consistency. It should be noted that the definitions in Δ^d below may contain s-predicate symbols on the right and hence need not be restricted.

THEOREM 1 (Definitional Replacement). *Suppose that Δ is a standardly complete theory. For $\mathbf{S}$ a set of s-predicate symbols from Δ , let Δ^c be the set of sentences of Δ containing no s-predicate symbols from $\mathbf{S}$ and let Δ' be a finite set of sentences of Δ . There is then a system of nonmodal definitions Δ^d of the s-predicate symbols of $\mathbf{S}$ in terms of the predicate symbols of $\mathcal{L}(\Delta) - \mathbf{S}$ for which $\Delta^c \cup \Delta^d$ subsumes Δ' .*

Proof. It suffices to prove this result for any finite subset Δ'' of $\Delta - \Delta^c$ for which $\Delta^c \cup \Delta''$ subsumes Δ' . For if $\Delta^c \cup \Delta^d$ subsumes Δ'' , it subsumes $\Delta^c \cup \Delta''$; and so, given that $\Delta^c \cup \Delta''$ subsumes Δ' , $\Delta^c \cup \Delta^d$ subsumes Δ' , by the transitivity of subsumption. Thus for the purposes of the proof, any sentence of Δ' may be replaced by a finite number of sentences from Δ that deductively imply it and any sentences of Δ' that belong to Δ^c may be dropped.

Let us now describe two ways in which this may be done. First, let $C_1, \dots, C_n$ be the truth-functional constituents that occur in the truth-functionally complex sentences B of Δ' (a truth-functional constituent being any sentence that is not, in primitive notation, either a negation or a disjunction). Then we may drop the sentences B of Δ' in favour of the subset Δ'' of the sentences $\pm C_1, \pm C_2, \dots, \pm C_n$ that belong to Δ ($\pm C_i$ is either C_i or its negation). For suppose $B \in \Delta'$. By Δ N-complete, either C_i or $\sim C_i$ belongs to Δ , for $i = 1, \dots, n$ (since each C_i is negatable). By truth-functional logic, $\Delta'' \vdash B$ or $\Delta'' \vdash \sim B$. But if $\Delta \vdash \sim B$, Δ is inconsistent. Thus, in either case, $\Delta'' \vdash B$; and hence Δ'' subsumes Δ' .

Second, suppose that Δ'' contains a sentence of the form $\sim F \subset G$. By Δ Q-complete, $(Fa \& \sim Ga) \in \Delta$ for some individual constant a . Let Δ''' be the result of replacing each such sentence $\sim F \subset G$ in Δ with the corresponding sentence $Fa \& \sim Ga$. Then clearly Δ''' subsumes Δ'' .

It therefore suffices to prove the result for $\Delta''' - \Delta^c$, which, for convenience, we redesignate as Δ' . It should be clear that Δ' consists of sentences of the following six kinds:

- (i) $R \subset S$, for $R, S \in \mathbf{S}$;
- (ii) $S \subset F$, for $S \in \mathbf{S}$ and F a predicate from $\mathcal{L}(\Delta) - \mathbf{S}$;
- (iii) $F \subset S$, for $S \in \mathbf{S}$ and F a predicate from $\mathcal{L}(\Delta) - \mathbf{S}$;
- (iv) Sa , for $S \in \mathbf{S}$;
- (v) $\sim Sa$, for $S \in \mathbf{S}$;
- (vi) $\Box_S B$, for $S \in \mathbf{S}$ and B an original sentence from $\mathcal{L}(\Delta) - \mathbf{S}$.

Note that the predicate F in (ii) and (iii) will either be an original predicate or an s-predicate symbol from $\mathcal{L}(\Delta) - \mathbf{S}$. The set Δ^d is defined as follows. Given any predicate symbol $S \in \mathbf{S}$, let $F_1, \dots, F_m$ be all of the closed predicates F from $\mathcal{L}(\Delta') - \mathbf{S}$ for which $\Delta' \vdash S \subset F$, and let

$a_1, \dots, a_n$ be all of the a in $\mathcal{L}(\Delta')$ for which $\Delta' \vdash \sim Sa$. We then let F_S be the predicate $\lambda x(F_1x \& \dots \& F_mx \& x \neq a_1 \& \dots \& x \neq a_n)$ and let Δ_d consist of the definitions $S \approx F_S$ for each $S \in \mathbf{S}$. Clearly, Δ^d is a system of definitions of the required sort. It therefore remains to show that $\Delta^c \cup \Delta^d$ subsumes Δ' .

Given a sentence B of Δ' , let B^* be the result of replacing each s -predicate symbol S of $\mathbf{S}$ in B with its definiens F_S . It then suffices to show:

(*) $\Delta' \vdash B^*$ for each sentence $B \in \Delta'$.

For then $\Delta \vdash B^*$ and so, given that B^* contains no predicate symbols of $\mathbf{S}$ and that Δ is a theory, it follows that $\Delta^c \vdash B^*$ and hence, using the definitions of Δ^d , that $\Delta^c \cup \Delta^d \vdash B$.

Let us first note that:

(**) $\Delta' \vdash S \subset F_S$ for each $S \in \mathbf{S}$.

For if F_ix is a conjunct within F_S , then $\Delta' \vdash S \subset F_i$ by definition; and if $x \neq a_j$ is a conjunct within F_S , then $\Delta' \vdash \sim Sa_j$ by definition and so $\Delta' \vdash \prod x(Sx \rightarrow x \neq a_j)$.

We now consider each sentence B in Δ' of the kinds (i)–(vi) in turn:

- (i) $B = R \subset S$ for $R, S \in \mathbf{S}$. We wish to show that $\Delta' \vdash B^*$, i.e. that $\Delta' \vdash F_R \subset F_S$. Suppose that F_ix is a conjunct within the predicate F_S . Then $\Delta' \vdash S \subset F_i$. Given that $\Delta' \vdash R \subset S$, $\Delta' \vdash R \subset F_i$; and so F_ix is also a conjunct within F_R . But then (a) $F_R \subset F_i$ is a theorem. Now suppose that $x \neq a_j$ is a conjunct within F_S . Then $\Delta' \vdash \sim Sa_j$. Given that $\Delta' \vdash R \subset S$, $\Delta' \vdash \sim Ra_j$; and so $x \neq a_j$ is also a conjunct within F_R . But then (b) $\prod x(F_Rx \rightarrow x \neq a_j)$ is a theorem. From (a) and (b), $F_R \subset F_S$ is a theorem and hence deducible from Δ' .
- (ii) $S \subset F$, for $S \in \mathbf{S}$ and F a closed predicate from $\mathcal{L}(\Delta') - \mathbf{S}$. Then Fx is a conjunct within F_S ; and so $F_S \subset F$ is a theorem and hence deducible from Δ' .
- (iii) $F \subset S$, for $S \in \mathbf{S}$ and F a closed predicate from $\mathcal{L}(\Delta') - \mathbf{S}$. By (**), $\Delta' \vdash S \subset F_S$. So since $F \subset S \in \Delta'$, $\Delta' \vdash F \subset F_S$, as required.
- (iv) Sa , for $S \in \mathbf{S}$. If F_ix is a conjunct within the predicate F_S , then $\Delta' \vdash S \subset F_i$ and so $\Delta' \vdash F_ia$. If $x \neq a_j$ is a conjunct within F_S , then $\Delta' \vdash \sim Sa_j$ and so $\Delta' \vdash a \neq a_j$.
- (v) $\sim Sa$, for $S \in \mathbf{S}$. Then $a = a_j$ for some j and so $\sim F_Sa$ is a theorem.

- (vi) $\Box_S A$, for $S \in \mathbf{S}$ and A an original sentence from $\mathcal{L}(\Delta)$. Given that $\Box_S A \in \Delta'$, it follows by $(**)$ and subsumption that $\Delta' \vdash \Box_{F_S} A$.

The restricted sentences were so chosen as to be adequate for establishing the relevant properties of the canonical model and yet susceptible to elimination in this way.

The first application of the theorem shows that, under suitable conditions, any restricted consequence of a theory can be represented as an consequence within the original language.

COROLLARY 2 (Substitution). *Let Δ_0 be a fully complete theory and Δ a safe and consistent extension of Δ_0 . Suppose $\Delta \vdash A = A(b_1, \dots, b_k, S_1, \dots, S_l)$, where $b_1, \dots, b_k$ are all of the distinct constants and $S_1, \dots, S_l$ all of the distinct s-predicate symbols that occur in A but not in Δ_0 , $k, l \geq 0$. There are then constants $a_1, \dots, a_k$ and s-predicate symbols $R_1, \dots, R_l$ from Δ_0 for which $\Delta_0 \vdash A(a_1, \dots, a_k, R_1, \dots, R_l)$.*

Proof. Given that $\Delta \vdash A$, $\Delta^+ \vdash A$ for some standardly complete extension Δ^+ of Δ ; and so we may suppose, without loss of generality, that Δ is already standardly complete. We may also suppose, without loss of generality, that $k, l > 0$.

Given that $\Delta \vdash A = A(b_1, \dots, b_k, S_1, \dots, S_l)$, there will be a finite subset Δ' of Δ for which $\Delta' \vdash A$. Let the $\mathbf{S}$ of the theorem be the set of s-predicate symbols of Δ . Thus Δ^c will be the set of original sentences of Δ , those that contain no s-predicate symbols. By the theorem, there will be a system of nonmodal definitions of the s-predicate symbols of Δ (in terms of the original predicate symbols of Δ and hence, by safety, of Δ_0) for which $\Delta^c \cup \Delta^d \vdash A$. So there will be sentences $C_1, \dots, C_m$ of Δ^c and sentences $D_1, \dots, D_n$ of Δ^d such that $C_1, \dots, C_m, D_1, \dots, D_n \vdash A$. We may suppose that the $D_1, \dots, D_n$ are of the form $S_1 \approx F_1, \dots, S_l \approx F_l$ for certain original predicates $F_1, \dots, F_l$, since any definition of the form $S \approx F$, where S is not one of $S_1, \dots, S_l$ may be eliminated by the Rule of Predicate Elimination; and by Δ^c a theory, we may suppose that there is a single sentence C from Δ^c for which:

- (a) $C, S_1 \approx F_1, \dots, S_l \approx F_l \vdash A$.

By substituting b_1 for all those constants of C and of $F_1, \dots, F_l$ that do not occur in A , we may also suppose that $b_1, \dots, b_k$ are the only constants occurring in C or any of the F_i , $i = 1, \dots, n$.

Let us write C in the form $C(b_1, \dots, b_k)$ and each F_i in the form $F_i(b_1, \dots, b_k)$. Then $\Delta \vdash C' = \Sigma x_1 \dots x_k C(x_1 \dots x_k)$. Since C' contains only original r-predicates and no constants, it belongs to the language of Δ_0 ; and since Δ is a theory, $C' \in \Delta_0$. Since Δ_0 is also Q-complete,

$C(a_1, \dots, a_k) \in \Delta_0$ for some constants $a_1, \dots, a_k$; and since Δ_0 is also P-complete, there are s-predicate symbols $R_i, i = 1, \dots, l$, for which $R_i \approx F_i(a_1, \dots, a_k) \in \Delta_0$. Thus $C(a_1, \dots, a_k), R_1 \approx F_1(a_1, \dots, a_k), \dots, R_l \approx F_l(a_1, \dots, a_k) \in \Delta_0$. But substituting $a_1, \dots, a_k$ for $b_1, \dots, b_k$ and $R_1, \dots, R_l$ for $S_1, \dots, S_l$ in (a), we obtain:

$$(b) C(a_1, \dots, a_k), R_1 \approx F_1(a_1, \dots, a_k), \dots, R_l \approx F_l(a_1, \dots, a_k) \vdash A(a_1, \dots, a_k, R_1, \dots, R_l),$$

and hence $\Delta_0 \vdash A(a_1, \dots, a_k, R_1, \dots, R_l)$, as required. $\square$

The next result shows that companionship is preserved under the addition of restricted sentences.

COROLLARY 3. *Suppose that Γ is a P-companion to the fully complete theory Δ and that Δ^+ is a consistent and safe extension of Δ that contains no constants or s-predicate symbols from Γ that are not already in Δ . Then Γ is also a P-companion to Δ^+ .*

Proof. Since Δ contains the r-predicate symbol P, so does Δ^+ . Suppose now that Γ is not a P-companion to Δ^+ . Then, for some sentence A of $\mathcal{L}(\Delta^+)$, $\Delta^+ \vdash \Box_P A$ and $\Gamma \vdash \sim A$. Write A in the form $A(b_1, \dots, b_k, S_1, \dots, S_l)$, where $b_1, \dots, b_k$ are all of the distinct constants and $S_1, \dots, S_l$ all of the distinct s-predicate symbols that occur in A but not in Δ . By the previous corollary, there are constants $a_1, \dots, a_k$ and s-predicate symbols $R_1, \dots, R_l$ of $\mathcal{L}(\Delta)$ for which $\Delta \vdash \Box_P A(a_1, \dots, a_k, R_1, \dots, R_l)$. Since $b_1, \dots, b_k, S_1, \dots, S_l$ do not occur in Γ , $\Gamma \vdash \sim A(a_1, \dots, a_k, R_1, \dots, R_l)$ and so Γ is not a P-companion to Δ . $\square$

6. COMPANIONSHIP

We establish some results that will help us construct suitable companions. We first establish three general results on the existence of companions and then show how companions may be extended to comply with various partial criteria for completeness.

We say that a constant a is *P-bound* in the set Δ if $a, P \in \mathcal{L}(\Delta)$ and $\Delta \vdash Pa$ and that an r-predicate symbol Q is *P-bound* in Δ if $Q, P \in \mathcal{L}(\Delta)$ and $\Delta \vdash Q \subset P$. We say that the set of sentences Γ is *P-bound* in Δ if each constant and r-predicate symbol of Γ is P-bound in Δ ; and we let $\mathcal{L}_P(\Delta) = \{a: a \text{ is P-bound in } \Delta\} \cup \{Q: Q \text{ is P-bound in } \Delta\}$. We may give similar definitions of $\sim P$ -bound but with $\sim Pa$ and $\sim(Q \subset P)$ taking the place of Pa and $Q \subset P$.

Given a set of sentences Δ , let $\Diamond_P \Delta = \{\Diamond_P(B_1 \& \dots \& B_n) : B_1, \dots, B_n \in \Delta\}$. Then for P-bound sets Γ , the criterion for companionship can be stated in terms of $\Diamond_P \Gamma$:

LEMMA 1. *Suppose that Γ is P-bound in Δ . Then Γ is a P-companion to Δ as long as $P \in \mathcal{L}(\Delta)$ and $\Delta \cup \Diamond_P \Gamma$ is consistent.*

Proof. Suppose $P \in \mathcal{L}(\Delta)$ but Γ is not a P-companion to Δ . Then for some sentence A of $\mathcal{L}(\Delta)$, $\Delta \vdash \Box_P A$ but $\Gamma \vdash \sim A$. Therefore, for some sentences $B_1, \dots, B_n$ of Γ , $B_1, \dots, B_n \vdash \sim A$. By truth-functional logic, $\vdash A \rightarrow \sim(B_1 \& \dots \& B_n)$; and so by Theorem 4(ii) of LE, $\vdash (\Box_P A \& |B_1| \subset P \& \dots \& |B_n| \subset P) \rightarrow \Box_P \sim(B_1 \& \dots \& B_n)$. Since Γ is P-bound by Δ , $\Delta \vdash |B_1| \subset P \& \dots \& |B_n| \subset P$. But then $\Delta \vdash \Box_P \sim(B_1 \& \dots \& B_n)$ and $\Delta \cup \{\Diamond_P(B_1 \& \dots \& B_n)\}$ is not consistent. $\square$

Given a sentence A and a set of constants $\mathbf{a}$, let $\Sigma \mathbf{a} A$ be $\Sigma x_1 \dots \Sigma x_n A'$, where A' is the result of replacing the distinct constants $a_1, \dots, a_n$ of $\mathbf{a}$ in A by distinct free variables $x_1, \dots, x_n$ (the choice of the free variables $x_1, \dots, x_n$ in this definition does not matter); and given a set of sentences Δ and a set of constants $\mathbf{a}$, let $\Sigma \mathbf{a} \Delta = \{\Sigma \mathbf{a}(A_1 \& \dots \& A_n) : A_1, \dots, A_n \in \Delta\}$. Then companionship for Γ can be stated in terms of companionship for $\Sigma \mathbf{a} \Gamma$:

LEMMA 2. *Let $\mathbf{a}$ be the set of constants of Γ that do not appear in Δ . Then Γ is a P-companion to Δ as long as $\Sigma \mathbf{a} \Gamma$ is a P-companion to Δ .*

Proof. Suppose that Γ is not a P-companion to Δ and that $P \in \mathcal{L}(\Delta)$. Then, for some sentence A of $\mathcal{L}(\Delta)$, $\Delta \vdash \Box_P A$ and $\Gamma \vdash \sim A$. So, for some sentences $B_1, \dots, B_n$ of Γ , $B_1, \dots, B_n \vdash \sim A$. Since the constants of $\mathbf{a}$ do not appear in A , it follows by quantificational reasoning that $\Sigma \mathbf{a}(B_1 \& \dots \& B_n) \vdash \sim A$. But then $\Sigma \mathbf{a} \Gamma$ is not a P-companion to Δ . $\square$

Putting together these two results, we obtain:

LEMMA 3. *Suppose that each r-predicate symbol of $\mathcal{L}(\Gamma)$ and each constant of $\mathcal{L}(\Gamma) \cap \mathcal{L}(\Delta)$ is P-bound in Δ . Let $\mathbf{a}$ be the set of constants of Γ that do not appear in Δ . Then Γ is a P-companion to Δ as long as $P \in \mathcal{L}(\Delta)$ and $\Delta \cup \Diamond_P \Sigma \mathbf{a} \Gamma$ is consistent.*

Proof. Suppose that $P \in \mathcal{L}(\Delta)$ and $\Delta \cup \Diamond_P \Sigma \mathbf{a} \Gamma$ is consistent. Under the given conditions, $\Sigma \mathbf{a} \Gamma$ is P-bound in Δ ; and so by Lemma 1, $\Sigma \mathbf{a} \Gamma$ is a P-companion to Δ . But $\mathbf{a}$ is the set of constants of Γ that do not appear in Δ ; and so, by Lemma 2, Γ is a P-companion to Δ . $\square$

We now show how to achieve partial compliance with N-completeness:

LEMMA 4. *Suppose that Γ is a P-companion to Δ and let A be any sentence. Then either $\Gamma \cup \{A\}$ or $\Gamma \cup \{\sim A\}$ is a P-companion to Δ .*

Proof. Suppose otherwise. Then, for some sentence B in $\mathcal{L}(\Delta)$, $\Delta \vdash \Box_p B$ and $\Gamma, A \vdash \sim B$ and, for some sentence C in $\mathcal{L}(\Delta)$, $\Delta \vdash \Box_p C$ and $\Gamma, \sim A \vdash \sim C$. Clearly, $(B \& C)$ is in $\mathcal{L}(\Delta)$ and so, by Theorem IV.4(iv) of LE, $\Delta \vdash \Box_p (B \& C)$. But $\Gamma \vdash (\sim B \vee \sim C)$; and so $\Gamma \cup \{B \& C\}$ is inconsistent and hence Γ is not a P-companion to Δ . $\square$

We turn to the demands of Q-completeness. There are two cases, according as to whether or not the witnessing constant is required to be P-bound. In the second case, we also add Pa to the complementary set Δ . This is done in order that we may later show that the companion relationship is reciprocated. It should be noted that the witnessed existential in the formulation of these lemmas is not required to be a restricted sentence.

LEMMA 5. *Suppose that $\Gamma \cup \{\Sigma x A(x)\}$ is a P-companion to Δ and that a is a constant not appearing in Δ , Γ or $A(x)$. Then $\Gamma \cup \{A(a)\}$ is a P-companion to Δ .*

Proof. Suppose otherwise. Then, for some sentence A' of $\mathcal{L}(\Delta)$ not containing a , $\Delta \vdash \Box_p A$ and $\Gamma, A(a) \vdash \sim A'$. Since the constant a does not appear in Γ , $A(x)$ or A' , it follows by quantificational reasoning that $\Gamma, \Sigma x A(x) \vdash \sim A'$ and so $\Gamma \cup \{\Sigma x A(x)\}$ is not a P-companion to Δ after all. $\square$

LEMMA 6. *Suppose that $\Gamma \cup \Sigma x (Px \& B(x))$ is a P-companion to Δ and that $\Delta \vdash \Sigma x Px$. Let a be a constant not appearing in Δ , Γ or $B(x)$. Then $\Gamma \cup \{Pa, B(a)\}$ is a P-companion to $\Delta \cup \{Pa\}$.*

Proof. Suppose otherwise. Then, for some sentence $A(a)$ of $\mathcal{L}(\Delta) \cup \{a\}$: $\Delta, Pa \vdash \Box_p A(a)$ and $\Gamma, Pa, B(a) \vdash \sim A(a)$. So $\Delta \vdash Pa \rightarrow \Box_p A(a)$; so since a does not appear in Δ , $\Delta \vdash Px \rightarrow \Box_p A(x)$; and hence $\Delta \vdash \prod x (Px \rightarrow \Box_p A(x))$. Since $\Delta \vdash \Sigma x Px$, it follows by Corollary V.2(i) of LE that $\Delta \vdash \Box_p \prod x (Px \rightarrow A(x))$. Since $\Gamma, Pa, B(a) \vdash \sim A(a)$ and a does not appear in Γ , $\Gamma \vdash \prod x (Px \rightarrow (A(x) \rightarrow \sim B(x)))$. But then $\Gamma, \prod x (Px \rightarrow A(x)) \vdash \prod x (Px \rightarrow \sim B(x))$. Therefore $\Gamma \cup \{\Sigma x (Px \& B(x)), \prod x (Px \rightarrow A(x))\}$ is inconsistent and so $\Gamma \cup \{\Sigma x (Px \& B(x))\}$ is not a P-companion to Δ . $\square$

We consider next compliance with P-completeness. There are five cases in all. The first is for when the witnessing predicate Q is not required to be P-bound. The others are for when it is. Just as in the case of the constants, reciprocation will require that the witnessing predicate in these four cases be P-bound on both left and right. The cases divide into two pairs. The first

of each pair deals with the special case in which application of definitional replacement is not required, and the second of each pair with the more general case in which it is. The first of the two pairs deals with the case in which witnessing is done on the ‘right’, in Γ , and the second of the two pairs with the case in which it is done on the ‘left’, in Δ . The significance of this distinction is that the left side may contain original r-predicates that do not appear on the right.

LEMMA 7. *Suppose that Γ is a P-companion to Δ and that Q is an s-predicate symbol that does not appear in Δ or Γ or F . Then $\Gamma \cup \{Q \approx F\}$ is a P-companion to Δ .*

Proof. Suppose otherwise. Then, for some sentence A of $\mathcal{L}(\Delta)$, $\Delta \vdash \Box_P A$ and $\Gamma, Q \approx F \vdash \sim A$. Since Q does not appear in Γ , F or A , it follows by Predicate Elimination that $\Gamma \vdash \sim A$; and so Γ is not a P-companion to Δ . $\square$

LEMMA 8. *Suppose that Γ is a P-companion to Δ , that each r-predicate symbol of $\mathcal{L}(\Gamma)$ and each constant of $\mathcal{L}(\Gamma) \cap \mathcal{L}(\Delta)$ is P-bound in Δ , and that $\Gamma \vdash F \subset P$ for some original closed predicate F of $\mathcal{L}(\Gamma)$. Let Q be an r-predicate symbol that does not appear in Δ , Γ or F . Then $\Gamma \cup \{Q \approx F\}$ is a P-companion to $\Delta \cup \{Q \subset P\}$.*

Proof. Suppose that $\Gamma \cup \{Q \approx F\}$ is not a P-companion to $\Delta \cup \{Q \subset P\}$. Let $\mathbf{a}$ be the set of constants of Γ that do not appear in Δ . Then by Lemma 3, $\Delta \cup \{Q \subset P\} \cup \Diamond_P \Sigma \mathbf{a}(\Gamma \cup \{F \approx Q\})$ is inconsistent. So for some finite subset $\Delta' = \{B_1, \dots, B_l\}$ of Δ and some finite subset $\Gamma' = \{C_1, \dots, C_m\}$ of Γ , $\Delta' \cup \{Q \subset P\} \cup \{\Diamond_P \Sigma \mathbf{a}(\Gamma' \cup \{Q \approx F\})\}$ is inconsistent. Let $a_1, \dots, a_m$ be the constants of $\mathbf{a}$ that appear in the members of Γ' or in F , and write each member C_i of Γ' in the form $C(a_1, \dots, a_m)$ and F in the form $F(a_1, \dots, a_m)$. Then $\{B_1, \dots, B_l\} \cup \{Q \subset P\} \cup \{\Diamond_P \Sigma x_1 \dots x_m (C_1(x_1, \dots, x_m) \& \dots \& C_n(x_1, \dots, x_m) \& Q \approx F(x_1, \dots, x_m))\}$ is inconsistent. Let $B = (B_1 \& \dots \& B_l)$ and $C(x_1, \dots, x_m) = (C_1(x_1, \dots, x_m) \& \dots \& C_n(x_1, \dots, x_m))$ and write the sequence of variables $x_1, \dots, x_m$ as $\mathbf{x}$. Then:

$$\vdash B \rightarrow (Q \subset P \rightarrow \Box_P \prod \mathbf{x} (Q \approx F(\mathbf{x}) \rightarrow \sim C(\mathbf{x})).$$

So by the modal rule of predicate elimination:

$$\vdash B \rightarrow \Box_P \prod \mathbf{x} (F(\mathbf{x}) \subset P \rightarrow \sim C(\mathbf{x})).$$

Hence $\Delta \vdash \Box_P \prod \mathbf{x} (F(\mathbf{x}) \subset P \rightarrow \sim C(\mathbf{x}))$. But $\Gamma \vdash F(a_1, \dots, a_m) \subset P$ and $\Gamma \vdash C(a_1, \dots, a_m)$. So the set $\{\prod x_1 \dots x_n (F(x_1, \dots, x_m) \subset P \rightarrow \sim C(x_1, \dots, x_m)), F(a_1, \dots, a_m) \subset P, C(a_1, \dots, a_m)\}$ is inconsistent and Γ is not a P-companion to Δ after all. $\square$

We may relax the requirement that each r-predicate symbol of Γ be P-bound in Δ :

LEMMA 9. *Suppose that the standardly complete theory Γ is a P-companion to Δ , that each original r-predicate symbol of Γ appears in Δ , that each constant and r-predicate symbol of $\mathcal{L}(\Delta) \cap \mathcal{L}(\Gamma)$ is P-bound in Δ , and that $\Gamma \vdash F \subset P$ for some original closed predicate F of $\mathcal{L}(\Gamma)$. Let Q be an r-predicate symbol that does not appear in Δ , Γ or F . Then $\Gamma \cup \{Q \approx F\}$ is a P-companion to $\Delta \cup \{Q \subset P\}$.*

Proof. Suppose that $\Gamma \cup \{Q \approx F\}$ is not a P-companion to $\Delta \cup \{Q \subset P\}$. Then, for some sentence A of $\mathcal{L}(\Delta) \cup \{Q\}$, $\Delta, Q \subset P \vdash \Box_p A$ and $\Gamma, Q \approx F \vdash \sim A$. So, for some finite subset Γ' of Γ , $\Gamma' \cup Q \approx F \vdash \sim A$. We now apply the theorem on definitional replacement (V.1) to Γ and Γ' , with S the set of s-predicate symbols of Γ that do not appear in Δ . Thus Γ^c is the subset of sentences from Γ that contain only s-predicate symbols from Δ ; and, since every original r-predicate symbol in Γ appears in Δ , every r-predicate symbol of Γ^c appears in Δ and hence is P-bound in Δ . By the theorem, there is a system of definitions Γ^d of the s-predicate symbols of S in terms of the s-predicate symbols of $\mathcal{L}(\Gamma) - S$ for which $\Gamma^c \cup \Gamma^d$ subsumes Γ' . Since $\Gamma', Q \approx F \vdash \sim A$, $\Gamma' \vdash Q \approx F \rightarrow \sim A$. So $\Gamma^c, \Gamma^d \vdash Q \approx F \rightarrow \sim A$; and so $\Gamma^c, \Gamma^d, Q \approx F \vdash \sim A$. Since F is an original predicate, none of the definienda of Γ^d appear in F and, by stipulation, none of them is Q . Moreover, none can appear in A , since A is in $\mathcal{L}(\Delta)$. So by successive applications of Predicate Elimination, $\Gamma^c, Q \approx F \vdash \sim A$.

We now apply the previous lemma to Γ^c and Δ . Every constant and r-predicate symbol of $\mathcal{L}(\Gamma) \cap \mathcal{L}(\Delta)$ is P-bound in Δ , and so every constant and r-predicate symbol of Γ^c is P-bound in Δ . Since Γ is a P-companion to Δ , $\Gamma^c \cup \{F \subset P\}$ is a P-companion to Δ and so, by the previous lemma, $\Gamma^c \cup \{Q \approx F\}$ is a P-companion to $\Delta \cup \{Q \subset P\}$. But this is contrary to the fact that $\Delta, Q \subset P \vdash \Box_p A$ and $\Gamma^c, Q \approx F \vdash \sim A$. $\square$

Without the theorem on Definitional Replacement (and the restriction in the class of sentences upon which it depends), it is hard to see how this crucial result might be established.

Let us now consider the pair of cases in which the predicate is witnessed on the left:

LEMMA 10. *Suppose that Γ is a P-companion to Δ , that each r-predicate symbol of $\mathcal{L}(\Gamma)$ and each constant of $\mathcal{L}(\Gamma) \cap \mathcal{L}(\Delta)$ is P-bound in Δ , and that $\Delta \vdash F \subset P$ for some original closed predicate F of $\mathcal{L}(\Delta)$. Let Q be an r-predicate symbol that does not appear in Δ , Γ or F . Then $\Gamma \cup \{Q \subset P\}$ is a P-companion to $\Delta \cup \{Q \approx F\}$.*

Proof. Suppose otherwise. Then, for some sentence A of $\mathcal{L}(\Delta)$, $\Delta \cup \{Q \approx F\} \vdash \Box_P A$ and $\Gamma, \{Q \subset P\} \vdash \sim A$. Let $\mathbf{a}$ be the set of constants of Γ that do not appear in Δ . Then by Lemma 3, $\Delta \cup \{Q \approx F\} \cup \Diamond_P \Sigma \mathbf{a}(\Gamma \cup \{Q \subset P\})$ is inconsistent. So, for some finite subset $\Gamma' = \{B_1, B_2, \dots, B_m\}$ of Γ , $\Delta \cup \{Q \approx F\} \cup \Diamond_P \Sigma \mathbf{a}(\Gamma' \cup \{Q \subset P\})$ is inconsistent. Let $a_1, a_2, \dots, a_n$ be the distinct constants of $\mathbf{a}$ that appear in the members of Γ' , and write each member B_i of Γ' in the form $B_i(a_1, \dots, a_n)$. Then $\Delta \cup \{Q \approx F\} \cup \{\Diamond_P \Sigma x_1, \dots, x_n (B_1(x_1, \dots, x_n) \& \dots \& B_m(x_1, \dots, x_n) \& (Q \subset P))\}$ is inconsistent. Let B abbreviate $\Sigma x_1 \dots x_n (B_1(x_1, \dots, x_n) \& \dots \& B_m(x_1, \dots, x_n))$. Then:

$$(*) \Delta, Q \approx F \vdash \Box_P ((Q \subset P) \rightarrow \sim B).$$

But $\Delta \vdash F \subset P$; so $\Delta, Q \approx F \vdash Q \subset P$; so by Theorem VI.2(i) of LE, $\Delta, Q \approx F \vdash \Box_P (Q \subset P)$; and so, given (*), $\Delta, Q \approx F \vdash \Box_P \sim B$. Since Q does not occur in F, B or Δ , it follows by Predicate Elimination that $\Delta \vdash \Box_P \sim B$. But given that $\Gamma \vdash (B_1(a_1, \dots, a_n) \& \dots \& B_m(a_1, \dots, a_n))$, $\Gamma \cup \{\sim B\}$ is inconsistent; and so Γ is not a P -companion to Δ after all. $\square$

LEMMA 11. *Suppose that the standardly complete theory Γ is a P -companion to Δ , that each original r -predicate symbol of Γ appears in Δ , that each constant and r -predicate symbol in $\mathcal{L}(\Delta) \cap \mathcal{L}(\Gamma)$ is P -bound in Δ , and that $\Delta \vdash F \subset P$ for some original closed predicate F of $\mathcal{L}(\Gamma)$. Let Q be an r -predicate symbol that does not appear in Δ, Γ or F . Then $\Gamma \cup \{Q \subset P\}$ is a P -companion to $\Delta \cup \{Q \approx F\}$.*

Proof. Obtained from Lemma 10 analogously to the way Lemma 9 is obtained from Lemma 8. $\square$

We turn finally to the question of reciprocation. Say that P is *closed* in Δ if $P \in \mathbf{L}(\Delta)$ and $\Delta \vdash \text{Cl}(P)$.

LEMMA 12 (Reciprocation). *Suppose that Γ is a broad P -companion to Δ , that every constant and r -predicate symbol P -bound in Γ is P -bound in Δ , that P is closed in either Δ or Γ , and that $P \in \mathcal{L}(\Gamma)$. Then Δ is a P -companion to Γ .*

Proof. Suppose that Δ is not a P -companion to Γ . Then, for some sentence A in $\mathcal{L}(\Gamma)$, $\Gamma \vdash \Box_P A$ and $\Delta \vdash \sim A$. So by modal reasoning:

$$(*) \Delta \vdash \Diamond_P \sim A.$$

Since $\Gamma \vdash \Box_P A$, $\Gamma \vdash |A| \subset cP$ from axiom (IV)(ii). Now it is readily shown that $\vdash \text{Cl}(P) \equiv (cP \subset P)$. So if $\Gamma \vdash \text{Cl}(P)$, then $\Gamma \vdash |A| \subset P$. Suppose, on the other hand, that $\Delta \vdash \text{Cl}(P)$ and hence that $\Delta \vdash cP \subset P$. From Theorem VI.2(i) of LE, it follows that $\Delta \vdash \Box_P (cP \subset P)$; and since

$\Gamma, cP \subset P \vdash |A| \subset P$, it follows from the fact that Γ is a broad companion that $\Gamma \vdash |A| \subset P$. Thus, in either case, $\Gamma \vdash |A| \subset P$. But then each constant and r-predicate symbol of A is P-bound in Γ and hence P-bound in Δ ; and so:

(**) $\Delta \vdash |A| \subset P$.

But from (*) and (**), it follows by Theorem IV.6(i) of LE that $\Delta \vdash \Box_P \Diamond_P \sim A$. Since $P \in \mathcal{L}(\Delta)$, $\Diamond_P A$ is in $\mathcal{L}(\Delta)$; and so, given that $\Gamma \vdash \Box_P A$, $\Gamma \cup \{\Diamond_P \sim A\}$ is inconsistent and Γ is not a P-companion to Δ . $\square$

Under the conditions of the lemma, Γ will be a P-companion to Δ and Δ to Γ . In this case, we say that they are (*reciprocal*) P-companions.

7. TWO-WORLD EQUILIBRIUM

As the first step towards constructing the canonical model, we show how to get two worlds in equilibrium, each a companion to the other and each fully complete. For simplicity, we describe the construction in three steps. First, we get an initial companion to be standardly complete; then we get it to be fully complete; and finally, we get both it and its complementary world to be fully complete.

Each step of the construction uses up infinitely many new constants and s-predicates and it is important always to leave behind infinitely many constants and s-predicates for subsequent use. This is a requirement that we shall not usually bother to state.

For the purposes of the induction, we need to impose some further requirements on the initial companions. Say that a set of sentences Δ is *P-determinate* if (i) P is in $\mathcal{L}(\Delta)$, (ii) for each constant a of $\mathcal{L}(\Delta)$, either $\Delta \vdash Pa$ or $\Delta \vdash \sim Pa$, and (iii) for each r-predicate symbol Q of $\mathcal{L}(\Delta)$, either $\Delta \vdash Q \subset P$ or $\Delta \vdash \sim(Q \subset P)$. Say that Δ and Γ are (*exactly*) *P-matching* if (i) Δ and Γ are P-determinate and (ii) $\mathcal{L}_P(\Delta) = \mathcal{L}_P(\Gamma) = \mathcal{L}(\Delta) \cap \mathcal{L}(\Gamma)$. We say that Γ is a *matching P-companion* to Δ if Γ is a P-companion to Δ and, in addition, Γ and Δ exactly P-match. For later purposes, we shall also say that Δ *almost P-matches* Γ if Δ and Γ are P-determinate and $\mathcal{L}_P(\Delta) \supseteq \mathcal{L}_P(\Gamma) = \mathcal{L}(\Delta) \cap \mathcal{L}(\Gamma)$ and that Γ is an *almost matching P-companion* to Δ if Γ is a P-companion to Δ and Δ almost P-matches Γ . We shall also say that Δ and Γ are an *almost matching pair* of P-companions if they are P-companions and either Δ almost P-matches Γ or Γ almost P-matches Δ . If Δ and Γ are an almost matching pair of P-companions, we shall say that Δ is *anomalous* if $\mathcal{L}_P(\Delta) \supsetneq \mathcal{L}_P(\Gamma)$. Thus

when the set Δ is anomalous it will contain some P-bound constants or predicate symbols that do not appear in Γ .

LEMMA 1. *Suppose that Γ is a matching P-companion to Δ . Then there are safe extensions Γ^+ of Γ and Δ^+ of Δ such that:*

- (i) Γ^* is a maximal matching P-companion to Δ^+ ; and
- (ii) Γ^+ is standardly complete.

Proof. Let $a_0, a_1, a_2, \dots$ be an infinite sequence of distinct constants not occurring in Δ or Γ (taking care to leave behind infinitely many constants from the whole language). Let $\mathcal{L}^+ = \mathcal{L}(\Gamma) \cup \{a_0, a_1, a_2, \dots\}$; and let $B_0, B_1, B_2, \dots$ be a repeating enumeration of the sentences of $\mathcal{L}^+$ (i.e. one in which each sentence occurs infinitely often). We define the sets Δ_n and Γ_n , for $n = 0, 1, 2, \dots$, by induction:

$n = 0$. $\Delta_0 = \Delta$ and $\Gamma_0 = \Gamma$.

$n = k + 1$. We first define the sets Δ'_k and Γ'_k :

- (a) If B_k is not in $\mathcal{L}(\Gamma_k)$ or $\Gamma_k \cup \{B_k\}$ is not a P-companion to Δ_k , then:

$$\Delta'_{k+1} = \Delta_k, \quad \text{and}$$

$$\Gamma'_{k+1} = \Gamma_k.$$

- (b) If B_k is in $\mathcal{L}(\Gamma_k)$ and $\Gamma_k \cup \{B_k\}$ is a P-companion to Δ_k , then:

$$\Delta'_{k+1} = \Delta_k, \quad \text{and}$$

$$\Gamma'_{k+1} = \Gamma_k \cup \{B_k\}.$$

We now define Γ_n and Δ_n :

- (c) If B_k is not in Γ'_k or is not of the form $\Sigma x C(x)$, then:

$$\Delta_{k+1} = \Delta'_{k+1}, \quad \text{and}$$

$$\Gamma_{k+1} = \Gamma'_{k+1}.$$

- (d) If $B_k \in \Gamma'_k$ it is of the form $\Sigma x C(x)$ and if $\Gamma'_{k+1} \cup \{\sim Pa, C(a)\}$ is a P-companion to Δ'_{k+1} , for a the first of the constants $a_0, a_1, a_2, \dots$ not to occur in Δ_k or Γ_k , then:

$$\Delta_{k+1} = \Delta'_{k+1}, \quad \text{and}$$

$$\Gamma_{k+1} = \Gamma'_{k+1} \cup \{\sim Pa, C(a)\}.$$

- (e) If $B_k \in \Gamma_k$ is of the form $\Sigma x C(x)$ and if $\Gamma'_{k+1} \cup \{\sim Pa, C(a)\}$ is *not* a P-companion to Δ'_{k+1} , for a the first of the constants $a_0, a_1, a_2, \dots$ not to occur in Δ_k or Γ_k , then:

$$\begin{aligned}\Delta_{k+1} &= \Delta'_{k+1} \cup \{Pa\} \\ \Gamma_{k+1} &= \Gamma'_{k+1} \cup \{Pa, C(a)\}.\end{aligned}$$

We prove by induction on n that Δ_n and Γ_n are matching P-companions:

$n = 0$. By supposition.

$n = k + 1$. We show that each of the operations specified under (a)–(e) above preserve the property of being a matching P-companion:

(a) Trivially true by IH since $\Delta'_{k+1} = \Delta_k$ and $\Gamma'_{k+1} = \Gamma_k$.

(b) By construction, $\Gamma'_{k+1} = \Gamma_k \cup \{B_k\}$ is a P-companion to $\Delta'_{k+1} = \Delta_k$. Γ'_{k+1} is P-determinate since $\Gamma'_{k+1} \supseteq \Gamma_k$, $\mathcal{L}(\Gamma'_{k+1}) = \mathcal{L}(\Gamma_k)$ and Γ_k is P-determinate by IH. Also $\mathcal{L}_P(\Gamma'_{k+1}) = \mathcal{L}_P(\Gamma_k)$. For if a (or Q) is P-bound in Γ'_{k+1} but not in Γ_k then $\sim Pa(\sim(Q \subset P)) \in \Gamma_k$ by Γ_k P-determinate, and so Γ'_{k+1} is inconsistent, contrary to its being a P-companion to Δ'_{k+1} .

(c) Trivial.

(d) By construction, Γ_{k+1} is a P-companion to $\Delta_{k+1} = \Delta'_{k+1}$. Since $\sim Pa \in \Gamma_{k+1}$, it follows by IH that Γ_{k+1} is P-determinate. Since $a \notin \mathcal{L}(\Delta_{k+1})$, $\mathcal{L}(\Delta_{k+1}) \cap \mathcal{L}(\Gamma_{k+1}) = \mathcal{L}(\Delta_k) \cap \mathcal{L}(\Gamma_k)$; since $\sim Pa \in \Gamma_{k+1}$, $\mathcal{L}_P(\Gamma_{k+1}) = \mathcal{L}_P(\Gamma_k)$, for if $Pa \in \Gamma_{k+1}$ then Γ_{k+1} is inconsistent.

(e) By IH, Γ'_k is a P-companion to Δ'_k . So by Lemma VI.4, either $\Gamma'_k \cup \{\Sigma x C(x), \Sigma x(C(x) \& \sim Px)\}$ or $\Gamma'_k \cup \{\Sigma x C(x), \sim \Sigma x(C(x) \& \sim Px)\}$ is a P-companion to Δ'_k . In the former case, $\Gamma'_k \cup \{C(a), \sim Pa\}$ is a P-companion to Δ'_k by Lemma VI.5; and since, by hypothesis, this is not so, the latter option holds. So $\Gamma'_k \cup \{\Sigma x(C(x) \& Px)\}$ is a P-companion to Δ'_k ; and so, by Lemma VI.6, $\Gamma_{k+1} = \Gamma'_{k+1} \cup \{Pa, Ca\}$ is a P-companion to $\Delta_{k+1} = \Delta'_{k+1} \cup \{Pa\}$. Since $Pa \in \Gamma_{k+1}$ and $Pa \in \Delta_{k+1}$, it follows by IH that both Δ_{k+1} and Γ_{k+1} are P-determinate. Clearly, $\mathcal{L}(\Delta_{k+1}) = \mathcal{L}(\Delta_k) \cup \{a\}$, $\mathcal{L}_P(\Delta_{k+1}) = \mathcal{L}_P(\Delta_k) \cup \{a\}$, $\mathcal{L}(\Gamma_{k+1}) = \mathcal{L}(\Gamma_k) \cup \{a\}$, and $\mathcal{L}_P(\Gamma_{k+1}) = \mathcal{L}_P(\Gamma_k) \cup \{a\}$; and therefore Γ'_{k+1} and Δ'_{k+1} exactly match.

Let $\Delta^+ = \Delta_0 \cup \Delta_1 \cup \dots$ and $\Gamma^+ = \Gamma_0 \cup \Gamma_1 \cup \dots$. Given that Γ_n is a matching P-companion to Δ_n for $n = 0, 1, \dots$, it is readily verified that Γ^+ is a matching P-companion to Δ^+ and that Γ^+ and Δ^+ contain no new original r-predicates. It remains to verify the various completeness requirements:

Maximality. Suppose that Γ^+ is not a maximal P-companion to Δ^+ . Then for some sentence B in $\mathcal{L}(\Gamma^+)$, $\Gamma^+ \cup \{B\}$ is a P-companion to Δ^+ but $B \notin \Gamma^+$. Since B is in $\mathcal{L}(\Gamma^+)$, B is in $\mathcal{L}(\Gamma_k)$ for some k ; and so $B = B_m$ for some $m \geq k$. But then the condition for (b) in the definition of Γ_{m+1} is satisfied and so $B = B_m \in \Gamma_{m+1} \subseteq \Gamma^+$ after all.

N-completeness. This follows from maximality. For given that Γ^+ is a P-companion to Δ^+ , it follows by Lemma VI.4 that either $\Gamma^+ \cup \{B\}$ or $\Gamma^+ \cup \{\sim B\}$ is a P-companion to Δ^+ for any negatable B in $\mathcal{L}(\Gamma^+)$. But then by the maximality of Γ^+ , either $B \in \Gamma^+$ or $\sim B \in \Gamma^+$.

Q-completeness. Suppose that $B = \Sigma x C(x) \in \Gamma^+$. Then $B \in \mathcal{L}(\Gamma_k)$ for some k . So $B = B_m$ for some $m \geq k$. But then the condition for (c) or (e) in the definition of Γ_{m+1} is satisfied and, in either case, $C(a) \in \Gamma_{m+1} \subseteq \Gamma^+$ for some constant a . $\square$

We now attempt to make the companion fully complete. Say that a set of sentences Δ is *P-restricted* (or that P is *restrictive on Δ*) if every original r -predicate in $\mathcal{L}(\Delta)$ is P -bound in Δ . Then:

LEMMA 2. *Suppose that Γ is a matching P -companion to Δ and that either Γ or Δ is P -restricted. Then there are safe extensions Δ^+ and Γ^+ of Δ and Γ which are such that (i) Γ^+ is a maximal matching P -companion to Δ^+ and (ii) Γ^+ is fully complete.*

Proof. Let $a_0, a_1, a_2, \dots$ be an infinite sequence of distinct constants not occurring in Δ or Γ and Q_0, Q_1, Q_2 an infinite sequence of distinct s -predicate symbols not occurring in Δ or Γ (taking care, as always, to leave behind infinitely many constants and s -predicate symbols from the language). Let $\mathcal{L}^+ = \mathcal{L}(\Gamma) \cup \{a_0, a_1, a_2, \dots\} \cup \{Q_0, Q_1, Q_2, \dots\}$; and let $F_0, F_1, F_2, \dots$ be a repeating enumeration of the original closed predicates of $\mathcal{L}^+$. We define the sets Δ_n and Γ_n , for $n = 0, 1, 2, \dots$, by induction:

$n = 0$. $\Delta_0 = \Delta$ and $\Gamma_0 = \Gamma$.

$n = k + 1, k \geq 0$. We first define the auxiliary sets Δ_k^+ and Γ_k^+ . To this end, we suppose that the set $\mathbf{A}$ of constants $\{a_0, a_1, a_2, \dots\}$ is partitioned into infinitely many infinite subsets $\mathbf{A}_0, \mathbf{A}_1, \mathbf{A}_2, \dots$. We then let Δ_k^+ and Γ_k^+ be safe extensions of Δ_k and Γ_k which conform to the conditions (i)–(ii) of the previous lemma and for which $\mathcal{L}(\Gamma_k^+) = \mathcal{L}(\Gamma_k) \cup \mathbf{A}_0 \cup \mathbf{A}_1 \cup \dots \cup \mathbf{A}_k$ if such extensions exist (otherwise we may, for the sake of definiteness, set $\Delta_k^+ = \Delta_k$ and $\Gamma_k^+ = \Gamma_k$).

We now define Δ_{k+1} and Γ_{k+1} :

- (a) if F_k is not in $\mathcal{L}(\Gamma_k^+)$ or if neither $\sim(F_k \subset P)$ nor $F_k \subset P$ belongs to Γ_k^+ , then:

$$\Delta_{k+1} = \Delta_k^+, \quad \text{and}$$

$$\Gamma_{k+1} = \Gamma_k^+.$$

- (b) If $\sim(F_k \subset P) \in \Gamma_k^+$ and if Q is the first of the s -predicate symbols in $Q_0, Q_1, Q_2, \dots$ not to appear in Δ_k^+ or Γ_k^+ , then:

$$\Delta_{k+1} = \Delta_k^+,$$

$$\Gamma_{k+1} = \Gamma_k^+ \cup \{Q \approx F_k\}.$$

- (c) If $(F_k \subset P) \in \Gamma_{k+1}$ and if Q is the first of the s -predicate symbols in $Q_0, Q_1, Q_2, \dots$ not to appear in Δ_k^+ or Γ_k^+ , then:

$$\Delta_{k+1} = \Delta_k^+ \cup \{Q \subset P\}, \quad \text{and}$$

$$\Gamma_{k+1} = \Gamma_k^+ \cup \{Q \approx F_k\}.$$

Suppose that Δ (resp. Γ) is P -restricted. We show by induction on n that Γ_n is a matching P -companion to Δ_n , that Γ_n (resp. Δ_n) is P -restricted, and that Γ_n and Δ_n are respective safe extensions of Γ and Δ .

$n = 0$. Trivial.

$n = k + 1$. By IH, Γ_k is a matching P -companion to Δ_k ; and so, by Lemma 1, Γ_k^+ will be a matching P -companion to Δ_k^+ . By IH, Γ_k (resp. Δ_k) is P -restricted. Since by Lemma 1, Γ_k^+ (Δ_k^+) is a safe extension of Γ_k (Δ_k), Γ_k^+ (Δ_k^+) is also P -restricted. Furthermore, since by IH, Γ_k and Δ_k are safe extensions of Γ and Δ , Γ_k^+ and Δ_k^+ are also safe extensions of Γ and Δ .

Let us now consider each of the three cases under the definition of Δ_n and Γ_n :

(a) Trivial.

(b) Γ_k^+ is a matching P -companion to Δ_k^+ ; and so, by Lemma VI.7, $\Gamma_{k+1} = \Gamma_k^+ \cup \{Q \approx F_k\}$ is a P -companion to $\Delta_{k+1} = \Delta_k^+ \cup \{Q \subset P\}$. Now Γ_k^+ P -matches Δ_k^+ . For since $Q \notin \mathcal{L}(\Delta_{k+1})$, $\mathcal{L}(\Delta_{k+1}) \cap \mathcal{L}(\Gamma_{k+1}) = \mathcal{L}(\Delta_k^+) \cap \mathcal{L}(\Gamma_k^+)$ and, since $\Gamma_{k+1} \vdash \sim(Q \subset P)$ and $Q \notin \mathcal{L}(\Delta_{k+1})$, Q is not P -bound in either Γ_{k+1} or Δ_{k+1} . Therefore $\mathcal{L}_P(\Gamma_{k+1}) = \mathcal{L}_P(\Gamma_k^+)$ and $\mathcal{L}_P(\Delta_{k+1}) = \mathcal{L}_P(\Delta_k^+)$ and so, by IH, $\mathcal{L}(\Delta_{k+1}) \cap \mathcal{L}(\Gamma_{k+1}) = \mathcal{L}_P(\Gamma_{k+1}) = \mathcal{L}_P(\Delta_{k+1})$. Moreover, Δ_k^+ and Γ_k^+ are P -determinate and so, since $\sim(Q \subset P) \in \Gamma_{k+1}$, Δ_{k+1} and Γ_{k+1} are P -determinate. The remaining conditions are readily verified.

(c) By IH, Γ_k^+ is a P -companion to Δ_k^+ . The case now divides according as to whether Δ_k^+ or Γ_k^+ is P -restricted:

(c') Γ_k^+ is P -restricted. It is easily seen that Γ_k^+ and Δ_k^+ conform to the conditions of Lemma VI.9 and so it follows from the lemma that $\Gamma_{k+1} = \Gamma_k^+ \cup \{Q \approx F_k\}$ is a P -companion to $\Delta_{k+1} = \Delta_k^+ \cup \{Q \subset P\}$. Since Q is P -bound in Δ_{k+1} and also in Γ_{k+1} (given $Q \approx F$, $F \subset P \in \Gamma$), Δ_{k+1} and Γ_{k+1} are P -matching. The remaining conditions are then readily verified.

(c'') Δ_k^+ is P -restricted. As for case (c') but using Lemma VI.11 in place of Lemma VI.9.

Let $\Delta^+ = \Delta_0 \cup \Delta_1 \cup \dots$ and $\Gamma^+ = \Gamma_0 \cup \Gamma_1 \cup \dots$. Given that Γ_n and Δ_n are matching P -companions for $n = 0, 1, \dots$, it is readily verified that Γ^+ and Δ^+ are matching P -companions; and likewise, given that Γ_n and Δ_n are safe extensions of Γ and Δ , it is evident that the same is true of

Γ^+ and Δ^+ . It remains to show that Γ^+ is fully complete and a maximal P-companion. Since Γ^+ is the union $\Gamma_0 \cup \Gamma_1 \cup \Gamma_2 \cup \dots$ of an increasing chain of sets $\Gamma_0 \subseteq \Gamma_1 \subseteq \Gamma_2 \subseteq \dots$ that are standardly complete, it is readily verified that Γ^+ itself is standardly complete. Likewise, given that each Γ_n , for $n = 1, 2, 3, \dots$ is a maximal P-companion to Δ_n and hence to Δ^+ , it is readily verified that Γ^+ is a maximal P-companion to Δ^+ .

It remains to establish P-completeness. Suppose that F is an original closed predicate in $\mathcal{L}(\Gamma^+)$. Then either $F \subset P$ or $\sim(F \subset P)$ belongs to Γ^+ by N-completeness; and so either $F \subset P$ or $\sim(F \subset P)$ belongs to Γ_k for some k . Now $F = F_m$ for some $m \geq k$; and so the condition for either (b) or (c) in the definition of Γ_{m+1} is satisfied. But in either case, $Q \approx F_m \in \Gamma_{m+1}$ for some s-predicate symbol Q .

We now get both sets to be fully complete:

LEMMA 3. *Suppose that Γ is a matching P-companion to Δ , with P either closed in Δ or restrictive on Γ . There are then safe extensions Δ^+ and Γ^+ of Δ and Γ which are maximal matching P-companions and fully complete.*

Proof. Clearly, by Lemma 2, we can suppose that Γ is a maximal P-companion to Δ and fully complete. We now define Δ_n and Γ_n by induction on $n = 0, 1, \dots$:

$n = 0$. Set $\Delta_0 = \Delta$ and $\Gamma_0 = \Gamma$.

$n = 2k + 1, k \geq 0$. Let Δ_{2k+1} and Γ_{2k+1} be safe extensions of Δ_{2k} and Γ_{2k} which are such that Δ_{2k+1} is fully complete and a maximal P-companion to Γ_{2k+1} (if such sets do not exist, let $\Delta_{2k+1} = \Delta_{2k}$ and $\Gamma_{2k+1} = \Gamma_{2k}$).

$n = 2k, k > 0$. Let Γ_{2k} and Δ_{2k} be safe extensions of Γ_{2k-1} and Δ_{2k-1} which are such Γ_{2k} is fully complete and a maximal P-companion to Δ_{2k} (if such sets do not exist, let $\Gamma_{2k} = \Gamma_{2k-1}$ and $\Delta_{2k} = \Delta_{2k-1}$).

We now show by induction on $n = 0, 1, 2, \dots$ that Γ_n and Δ_n are safe extensions of Γ and Δ and that Γ_n is a maximal matching and fully complete P-companion to Δ_n for even n and that Δ_n is a maximal matching and fully complete P-companion to Γ_n for odd n .

$n = 0$. Trivial.

$n = 2k + 1$. By IH, Γ_{2k} and Δ_{2k} are safe extensions of Γ and Δ and Γ_{2k} is a fully complete maximal matching P-companion to Δ_{2k} . We now apply the Reciprocation Lemma (VI.12) to Γ_{2k} and Δ_{2k} . Since Γ_{2k} is a maximal matching P-companion to Δ_{2k} , it is a broad P-companion to Δ_{2k} ; since Γ_{2k} and Δ_{2k} exactly P-match, every constant and r-predicate symbol P-bound in Γ_{2k} is P-bound in Δ_{2k} and, also, $P \in \mathcal{L}(\Gamma_{2k})$; and since Δ or Γ is P-closed, Δ_{2k} or Γ_{2k} is P-closed. The conditions for the application of

Lemma VI.12 are therefore satisfied and so we may conclude that Δ_{2k} is a P-companion to Γ_{2k} .

We now apply Lemma 2. Since Γ_{2k} is a *matching* P-companion to Δ_{2k} , Δ_{2k} is a *matching* P-companion to Γ_{2k} . Since Δ or Γ is P-restricted and Δ_{2k} and Γ_{2k} are safe extensions of Δ or Γ , either Δ_{2k} or Γ_{2k} is P-restricted. The conditions for the application of the lemma are therefore satisfied and so may conclude that there exist safe extensions Δ_{2k+1} and Γ_{2k+1} of Δ_{2k} and Γ_{2k} (and hence of Δ and Γ) with Δ_{2k+1} a maximal matching and fully complete P-companion to Γ_{2k+1} .

$n = 2k$. Similar to the previous case, but with left and right reversed.

Let $\Delta^+ = \Delta_0 \cup \Delta_1 \cup \dots$ and $\Gamma^+ = \Gamma_0 \cup \Gamma_1 \cup \dots$. Since Δ^+ is the union $\Delta_1 \cup \Delta_3 \cup \Delta_5 \cup \dots$ of an increasing chain $\Delta_1 \subseteq \Delta_3 \subseteq \Delta_5 \subseteq \dots$ of fully complete sets, it is fully complete. Likewise, since Γ^+ is the union $\Gamma_0 \cup \Gamma_2 \cup \Gamma_4 \cup \dots$ of an increasing chain $\Gamma_0 \subseteq \Gamma_2 \subseteq \Gamma_4 \subseteq \dots$ of fully complete sets, it is also fully complete. Furthermore, since Δ_n is a matching P-companion to Γ_n for $n = 1, 3, 5, \dots$, it is readily verified that Δ^+ is a matching P-companion to Γ^+ ; and, similarly, Γ^+ is a matching P-companion to Δ^+ . Finally, Δ_n and Γ_n are safe extensions of Δ and Γ and hence so are Δ^+ and Γ^+ .

We conclude this section with two results that show how the initial conditions for the previous constructions can be met.

LEMMA 4. *Suppose that Δ is a consistent set of sentences that yields both $\Diamond_P B$ and $|B| \subset P$, for $P \in \mathcal{L}(\Delta)$. Then $\Gamma = \{B\}$ is a P-companion to Δ .*

Proof. Suppose otherwise. Then there is a sentence A in $\mathcal{L}(\Delta)$ such that $\Delta \vdash \Box_P A$ and $B \vdash \sim A$. So $\vdash A \rightarrow \sim B$. But then by Theorem IV.4(vi) of LE, $\vdash \Box_P A \& (|B| \subset P) \rightarrow \Box_P \sim B$; and so Δ is inconsistent. $\square$

Let us say that P is *effectively closed* in Δ if whenever $\Delta \vdash \Box_P B$ for B in $\mathcal{L}(\Delta)$ and a (resp. Q) belongs to B , then Pa (resp. $Q \subset P$) is deducible from Δ . Clearly if P is closed in Δ then it is effectively closed in Δ . However, the converse need not hold even when Δ is N-complete, since $Cl(P)$ is not in general a restricted sentence.

LEMMA 5. *Suppose that P is effectively closed in Δ , with Δ P-determinate, that Γ is a P-companion to Δ , and that $\mathcal{L}(\Gamma) \cap \mathcal{L}(\Delta) \subseteq \mathcal{L}_P(\Delta)$ and $\mathcal{L}(\Gamma) - \mathcal{L}(\Delta) \subseteq \mathcal{L}_{\sim P}(\Gamma)$. Let $\Gamma^+ = \Gamma \cup \{B : \Delta \vdash \Box_P B, B \text{ in } \mathcal{L}(\Delta)\}$. Then Γ^+ is a matching P-companion to Δ .*

Proof. It is readily seen that Γ^+ is a P-companion to Δ . Let $\Delta_P = \{B : \Delta \vdash \Box_P B, B \text{ in } \mathcal{L}(\Delta)\}$. To verify that Γ^+ and Δ exactly match, we first establish the following identities:

$$(*) \mathcal{L}_P(\Delta) = \mathcal{L}(\Delta_P) = \mathcal{L}_P(\Gamma^+).$$

We merely consider the case of the constants of a language, since the proof for the case of r -predicates is similar. Note first that $\mathcal{L}_P(\Delta)$ is a subset of both $\mathcal{L}(\Delta_P)$ and $\mathcal{L}_P(\Gamma^+)$. For suppose $a \in \mathcal{L}_P(\Delta)$. Then $\Delta \vdash Pa$. But then $\Delta \vdash \Box_P Pa$; and since $a, P \in \mathcal{L}(\Delta)$, a belongs to both $\mathcal{L}(\Delta_P)$ and $\mathcal{L}_P(\Gamma^+)$. Now suppose $a \in \mathcal{L}(\Delta_P)$. Then $\Delta \vdash \Box_P B$ for some sentence B in $\mathcal{L}(\Delta)$ that contains a . Since P is effectively closed in Δ , $\Delta \vdash Pa$; and so $a \in \mathcal{L}_P(\Delta)$. Thus $\mathcal{L}_P(\Delta) = \mathcal{L}(\Delta_P)$. Finally, let us establish that $\mathcal{L}_P(\Gamma^+) \subseteq \mathcal{L}(\Delta_P)$. Suppose $a \in \mathcal{L}_P(\Gamma^+)$. Then either $a \in \mathcal{L}(\Delta_P)$ or $a \in \mathcal{L}(\Gamma)$. In the former case, there is nothing to prove. In the latter case, either $a \in \mathcal{L}(\Delta)$ or $a \notin \mathcal{L}(\Delta)$. In the former subcase, $a \in \mathcal{L}_P(\Delta)$ since $\mathcal{L}(\Gamma) \cap \mathcal{L}(\Delta) \subseteq \mathcal{L}_P(\Delta)$, and so $a \in \mathcal{L}(\Delta_P)$ by the previously proved identity. In the latter case, $a \in \mathcal{L}_{\sim P}(\Gamma)$ since $\mathcal{L}(\Gamma) - \mathcal{L}(\Delta) \subseteq \mathcal{L}_{\sim P}(\Gamma) = \mathcal{L}$. But then Γ^+ is inconsistent and cannot be a P -companion to Δ .

We now show that Γ^+ is P -determinate (again ignoring the case of predicate symbols). Suppose $a \in \mathcal{L}(\Gamma^+)$. Then either $a \in \mathcal{L}(\Delta_P)$ or $a \in \mathcal{L}(\Gamma) - \mathcal{L}(\Delta_P)$. In the former case, $a \in \mathcal{L}_P(\Delta)$ by (*). But then $\Delta \vdash \Box_P Pa$ and so $Pa \in \Delta_P \subseteq \Gamma^+$. In the latter case, $a \in \mathcal{L}(\Gamma) - \mathcal{L}_P(\Gamma)$ by (*) and, since $\mathcal{L}(\Gamma) \cap \mathcal{L}(\Delta) \subseteq \mathcal{L}_P(\Delta)$, $a \notin \mathcal{L}(\Delta)$ and so, given $\mathcal{L}(\Gamma) - \mathcal{L}(\Delta) \subseteq \mathcal{L}_{\sim P}(\Gamma)$, $\Gamma \vdash \sim Pa$. Furthermore, $\mathcal{L}_P(\Gamma^+) = \mathcal{L}_P(\Delta)$ by (*) and:

$$\begin{aligned} \mathcal{L}(\Delta) \cap \mathcal{L}(\Gamma^+) &= \mathcal{L}(\Delta) \cap (\mathcal{L}(\Gamma) \cup \mathcal{L}(\Delta_P)) \\ &= \mathcal{L}(\Delta) \cap (\mathcal{L}(\Gamma) \cup \mathcal{L}_P(\Delta)) \quad (\text{by } *) \\ &= (\mathcal{L}(\Delta) \cap \mathcal{L}(\Gamma)) \cup (\mathcal{L}(\Delta) \cap \mathcal{L}_P(\Delta)) \\ &= \mathcal{L}_P(\Delta) \quad \text{since } \mathcal{L}(\Delta) \cap \mathcal{L}(\Gamma) \subseteq \mathcal{L}_P(\Delta). \end{aligned}$$

And therefore Γ^+ and Δ exactly match. $\square$

We conclude with a result concerning a single set of sentences:

LEMMA 6. *Any consistent set of sentences Γ (containing only original predicate symbols and constants) has a safe and fully complete extension Γ^+ .*

Proof. The result can be proved directly using classical techniques to ensure standard completeness and the rule of predicate elimination to ensure P -completeness. Alternatively, we can add $P \approx \bigvee$ to Δ^+ , let $\Delta = \emptyset$, and apply Lemmas 5 and 2. $\square$

8. GLOBAL EQUILIBRIUM

We show how infinitely many worlds can be put into ‘equilibrium’ in such a way that all stated possibilities are realized. The worlds will be arranged in a tree with the arrows from one node to another representing the possibilities. Thus when a P-possibility at one node is realized by another node, there will be a ‘P-arrow’ in the tree linking the one node to the other. We must show how more and more arrows can be added to the tree without disturbing the equilibrium of the tree as a whole.

We begin with some terminology concerning trees. We suppose given an infinite number of ‘world constants’ $w_0, w_1, w_2, \dots$. A *diagram* $\mathcal{D}$ is then a set consisting of:

- (i) pairs $\langle w, B \rangle$, for w a world constant and B a (restricted) sentence;
- (ii) triples $\langle w, v, P \rangle$, for w and v distinct world-constants and P an s-predicate symbol.

We shall suppose that for any given world-constants w and v there is at most one P for which $\langle w, v, P \rangle \in \mathcal{D}$.

Given a diagram $\mathcal{D}$, we let:

- (i) $W_{\mathcal{D}} = \{w : w \text{ is component of an ordered pair or triple in } \mathcal{D}\}$;
- (ii) $\mathcal{D}_w = \{B : \langle w, B \rangle \in \mathcal{D}\}$ for any $w \in W_{\mathcal{D}}$;
- (iii) $\mathcal{L}(\mathcal{D}) = \bigcup \{\mathcal{L}(\mathcal{D}_w) : w \in W_{\mathcal{D}}\}$;
- (iv) $\ll_{\mathcal{D}} = \{\langle w, v \rangle : \langle w, v, P \rangle \in \mathcal{D}\}$.

When $\langle w, v, P \rangle \in \mathcal{D}$, we shall say that w is *linked to* v (by P) in $\mathcal{D}$. We call the pair $\langle w, v \rangle$ an *arrow in* $\mathcal{D}$ if $\langle w, v, P \rangle \in \mathcal{D}$ for some P and we call the pair $\langle w, v \rangle$ a *link in* $\mathcal{D}$ if either $\langle w, v \rangle$ or $\langle v, w \rangle$ is an arrow.

Given a diagram $\mathcal{D}$, let $T_{\mathcal{D}}$ be $\langle W_{\mathcal{D}}, \ll_{\mathcal{D}} \rangle$. $\mathcal{D}$ is then said to be a *tree diagram* if $T_{\mathcal{D}}$ is a tree. Suppose that w and v are two distinct nodes on a tree $T_{\mathcal{D}}$. Then by the *path from* w to v is meant the sequence of distinct nodes $w = w_0, w_1, \dots, w_n = v$ such that, for $i = 0, 1, \dots, n-1$, $\langle w_i, w_{i+1} \rangle$ is a link in the tree. Clearly, there is a unique path from any node of a tree $T_{\mathcal{D}}$ to any other node. A *subtree* T' of a tree T is a restriction of T in which paths are preserved (any path of T' is a path of T). If T is the tree of a diagram $\mathcal{D}$, then a subdiagram $\mathcal{D}'$ of $\mathcal{D}$ may be associated with a subtree of T in the obvious way.

We distinguish four types of extensions of a diagram. The tree diagram $\mathcal{D}^+$ is said to be an *extension* (simpliciter) of the tree diagram $\mathcal{D}$ if (i) $T_{\mathcal{D}}$ is a subtree of $T_{\mathcal{D}^+}$ and (ii) for every $w \in W_{\mathcal{D}}$, $\mathcal{D}_w \subseteq \mathcal{D}_w^+$; $\mathcal{D}^+$ is a *conservative extension* of $\mathcal{D}$ if, in addition, $T_{\mathcal{D}^+} = T_{\mathcal{D}}$; $\mathcal{D}^+$ is a *safe extension* of $\mathcal{D}$ if it contains no original r-predicate symbols not already in $\mathcal{D}$; and it is a *normal extension* of $\mathcal{D}$ if it is both safe and conservative.

Let us now specify some of the desirable properties of diagrams. Their presence will help us in constructing a ‘canonical’ model from the diagram. Suppose that $\langle w, v, P \rangle \in \mathcal{D}$. Then the arrow $\langle w, v \rangle$ in $\mathcal{D}$ is said to be *regular* if $\mathcal{D}_w$ and $\mathcal{D}_v$ are a maximal matching pair of fully complete P-companions with P closed in $\mathcal{D}_w$ and restrictive on $\mathcal{D}_v$. The arrow is said to be *almost regular* if $\mathcal{D}_w$ and $\mathcal{D}_v$ are an almost matching pair of fully complete P-companions for which P is closed in $\mathcal{D}_w$ and $\mathcal{D}_v$ is P-restricted. If an arrow $\langle w, v \rangle$ of $\mathcal{D}$ is almost regular, then w (v) is said to be its *anomalous* node if $\mathcal{D}_w$ (resp. $\mathcal{D}_v$) is the anomalous member of the pair $\mathcal{D}_w, \mathcal{D}_v$.

A tree diagram $\mathcal{D}$ itself is said to be *regular* if either it has a single node w for which $\mathcal{D}_w$ is fully complete or it has more than one node and:

- (i) each arrow in the diagram is regular; and
- (ii) if $w = w_0, w_1, \dots, w_n = v$ is a path from w to v and a (or Q) $\in \mathcal{L}(\mathcal{D}_w) \cap \mathcal{L}(\mathcal{D}_v)$, then a (resp. Q) $\in \mathcal{D}_{w_i}$ for $i = 1, \dots, n - 1$.

We call (ii) the path condition and if a (resp. Q) belongs to each member of the path $p = w_0, w_1, \dots, w_n$, then we call p an *a-path* (resp. *Q-path*).

The diagram is said to be *almost regular* under the same conditions but with (i) replaced by:

- (i') each arrow in the diagram is almost regular and there is at most one anomalous node in the diagram.

Let us explain how this terminology arises in the construction of the canonical model. Suppose that the sentence $\Diamond_P B \in \mathcal{D}_w$. Then we may wish to construct a world v that is linked to w by P and for which $B \in \mathcal{D}_v$. By previous results we can ensure that $\mathcal{D}_w$ and $\mathcal{D}_v$ exactly match. But we may now have another sentence $\Diamond_Q C \in \mathcal{D}_w$ and this may require us to construct another world u linked to w by Q . However, in getting $\mathcal{D}_w$ and $\mathcal{D}_u$ to exactly match, we will lose the exact match between $\mathcal{D}_w$ and $\mathcal{D}_v$. They will almost match, with w becoming an anomalous node.

We shall show how these anomalies can be ironed out. But first we need a result on the behaviour of a world-language under extensions:

To show that a diagram can retain regularity when we add new possibilities, we shall need to split the resulting diagram into two. Given a tree T , we say that $\langle T_1, T_2, w \rangle$ is a *splitting* of T if T_1 and T_2 are subtrees of T whose only node in common is w and which exhaust the nodes in T . Likewise, $\langle \mathcal{D}_1, \mathcal{D}_2, w \rangle$ is said to be a splitting of a tree diagram $\mathcal{D}$ if, for some T_1 and T_2 , $\langle T_1, T_2, w \rangle$ is a splitting of $T_{\mathcal{D}}$ and $\mathcal{D}_1$ and $\mathcal{D}_2$ are the respective restrictions of $\mathcal{D}$ to T_1 and T_2 . We show how one half of a splitting can be regularized without harm to the other:

LEMMA 1. Suppose that $\mathcal{D}$ is an almost regular tree-diagram and that $\langle \mathfrak{E}, \mathfrak{F}, w_0 \rangle$ is a splitting of $\mathcal{D}$ in which $\mathfrak{F}$ contains the sole anomalous node (if any) of $\mathcal{D}$. Let $\mathfrak{F}^+$ be a normal extension of $\mathfrak{F}$ which contains no constants or s-predicate symbols that occur in $\mathfrak{E}$ but not in $\mathfrak{F}$. Then the diagram $\mathcal{D}^+ = \mathfrak{E} \cup \mathfrak{F}^+$ is almost regular with w_0 as its sole anomalous node.

Proof. Let us verify the various conditions required for $\mathcal{D}^+$ to be almost regular:

First, take any w in $W_{\mathcal{D}}$. If w is in $W_{\mathfrak{F}}$, then $\mathcal{D}_w^+ = \mathfrak{F}_w^+$ is fully complete since $\mathfrak{F}^+$ is regular. If w is in $W_{\mathfrak{E}} - \{w_0\}$, then $\mathcal{D}_w^+ = \mathfrak{E}_w$ is fully complete since $\mathfrak{E}$ is almost regular.

Now take any $\langle w, v, P \rangle \in \mathcal{D}^+$. Then either $\langle w, v, P \rangle \in \mathfrak{E}$ or $\langle w, v, P \rangle \in \mathfrak{F}$. In either case, since $\mathfrak{E}$ and $\mathfrak{F}$ are both almost regular, P is closed in $\mathcal{D}_w$ and restrictive in $\mathcal{D}_v$. Given that P is closed in $\mathcal{D}_w$, it is closed in $\mathcal{D}_w^+ \supseteq \mathcal{D}_w$. Given that $\mathcal{D}_v$ is P -restricted and that $\mathcal{D}^+$ contains no original r-predicate symbols that do not occur in $\mathcal{D}$, $\mathcal{D}_v^+$ is also P -restricted.

Next, suppose $\langle w, v \rangle$ is either an arrow in $\mathfrak{F}$ or an arrow of $\mathfrak{E}$ for which neither w nor v is w_0 . In the first case, $\langle w, v \rangle$ is a regular arrow of $\mathcal{D}^+$ since $\mathfrak{F}^+$ is regular and both $\mathcal{D}_w^+ = \mathfrak{F}_w^+$ and $\mathcal{D}_v^+ = \mathfrak{F}_v^+$; and in the second case, $\langle w, v \rangle$ is a regular arrow of $\mathcal{D}^+$ since $\mathfrak{E}$ is almost regular, neither w nor v is anomalous and both $\mathcal{D}_w^+ = \mathfrak{E}_w$ and $\mathcal{D}_v^+ = \mathfrak{E}_v$.

In the remaining case, $\langle w, v, P \rangle \in \mathfrak{E}$ with either $w = w_0$ or $v = w_0$. Suppose that $w = w_0$ (the case in which $v = w_0$ is entirely similar). We must then show that $\langle w, v \rangle$ is an almost regular arrow in $\mathcal{D}^+$ with $w = w_0$ the anomalous node.

Since $\mathfrak{F}$ contains the sole anomalous node of $\mathcal{D}$, that node must be $w = w_0$. So, given that $\mathfrak{E}$ is almost regular, $\mathfrak{E}_w$ almost matches $\mathfrak{E}_v$. But then the only way $\mathcal{D}_w^+ = \mathfrak{F}_w^+$ could fail to almost match $\mathcal{D}_v^+ = \mathfrak{E}_v$ is if $\mathfrak{F}_w^+$ contained a constant or r-predicate symbol s that belonged to $\mathfrak{E}_v$ but not to $\mathcal{D}_w$. If s is new to $\mathfrak{F}^+$ then it cannot belong to $\mathfrak{E}$ by supposition. If s is in $\mathfrak{F}$, then it is in $\mathfrak{F}_u$ for some u . But there is a path from u to v that goes through w ; and so, by the application of the path condition to $\mathcal{D}$, s is in $\mathcal{D}_w$ after all.

Since $\mathcal{D}$ is almost regular, $\Gamma = \mathcal{D}_v$ is a P -companion to $\Delta = \mathcal{D}_w$. Letting $\Delta^+ = \mathcal{D}_w^+$, it is clear that the conditions for the application of Corollary V.3 are satisfied; and so $\mathcal{D}_v^+ = \mathcal{D}_v$ is a P -companion to $\mathcal{D}_w^+$. Now letting $\Gamma = \mathcal{D}_v^+$ and $\Delta = \mathcal{D}_w^+$, it is clear that the conditions for the application of the Reciprocation Lemma (VI.12) will be satisfied; for Γ is a maximal P -companion to $\mathcal{D}_w$, hence a maximal P -companion to Δ , and so a broad P -companion to Δ . Therefore, by the lemma, $\mathcal{D}_w^+$ is a P -companion to $\mathcal{D}_v^+$.

Finally, we must show that $\mathfrak{D}^+$ conforms to the path condition. The only difficulty concerns the case in which there is a path from v in $\mathfrak{E}$ to u in $\mathfrak{F}$ (where neither v nor u is w_0) with the constant or r -predicate symbol s in both $\mathfrak{D}_v^+ = \mathfrak{E}_v$ and $\mathfrak{D}_u^+ = \mathfrak{F}_u$. But then s already occurs in $\mathfrak{F}$ and so belongs to some $\mathfrak{F}_t$. But any path from v to t will go through w_0 and so, by the application of the path condition to $\mathfrak{D}$, s belongs to $\mathfrak{D}_w$ and hence to $\mathfrak{D}_w^+$. $\square$

LEMMA 2. *Any finite almost regular diagram $\mathfrak{D}$ has a normal regular extension $\mathfrak{D}^+$.*

Proof. By induction on the number $m \geq 1$ of nodes in $\mathfrak{D}$. If $\mathfrak{D}$ is already regular, there is nothing to prove. So we may assume that $\mathfrak{D}$ contains exactly one anomalous node w_1 .

$m = 1$. Trivial, since the notions of regular and almost regular coincide for diagrams with one node.

$m = 2$. Let the two worlds be w and v and suppose that $\langle w, v, P \rangle \in \mathfrak{D}$ with v not anomalous. We first apply Lemma VII.5. Clearly, P is closed in $\mathfrak{D}_w$. Since $\mathfrak{D}_w$ is N -complete, it is P -determinate. $\mathfrak{D}_v$ is a P -companion to $\mathfrak{D}_w$ by supposition. $\mathcal{L}(\Gamma) \cap \mathcal{L}(\Delta) \subseteq \mathcal{L}_P(\Delta)$, since Δ almost matches. Also $\mathcal{L}(\Gamma) - \mathcal{L}(\Delta) \subseteq \mathcal{L}_{\sim P}(\Delta)$ since if a constant or r -predicate symbol is in $\mathcal{L}(\Gamma) - \mathcal{L}(\Delta)$, then it is not P -bound in Γ and so, by Γ N -complete, it is $\sim P$ -bound in Γ . Since the condition for its application are met, it follows from the lemma that $(\mathfrak{D}_v)^+ = \mathfrak{D}_v \cup \{B : \mathfrak{D}_w \vdash \Box_P B, B \text{ in } \mathcal{L}(\mathfrak{D}_w)\}$ is a matching P -companion to $\mathfrak{D}_w$. So by Lemma VII.3, there are safe and fully complete extensions Γ^+ and Δ^+ of $\mathfrak{D}_w$ and $(\mathfrak{D}_v)^+$ which are maximal matching P -companions and fully complete. Let $\mathfrak{D}^+$ be the conservative extension of $\mathfrak{D}$ for which $\mathfrak{D}_w^+ = \Delta^+$ and $\mathfrak{D}_v^+ = \Gamma^+$. Then clearly $\mathfrak{D}^+$ is a regular tree-diagram.

The proof in case v is anomalous is similar but with the role of w and v reversed. In this case, P may not be closed in $\mathfrak{D}_v$ but it will be effectively closed. For suppose $\mathfrak{D}_v \vdash \Box_P B$ for B in $\mathcal{L}(\mathfrak{D}_v)$ and that a is a constant of B . If $Pa \notin \mathfrak{D}_v$, then $\sim Pa \in \mathfrak{D}_v$ by $\mathfrak{D}_v$ N -complete. But since P is closed in $\mathfrak{D}_w$, $\mathfrak{D}_w \vdash \Box_P Cl(P)$; and so $\mathfrak{D}_v \cup \{Cl(P)\}$ is inconsistent and $\mathfrak{D}_v$ is not a P -companion to $\mathfrak{D}_w$ after all.

$m > 2$. It should be clear that there is a splitting $\langle T_1, T_2, w_0 \rangle$ of $T_{\mathfrak{D}}$ in which neither T_1 nor T_2 consists of a single node. For if $T_{\mathfrak{D}}$ takes the form of a single branch, then we may split the tree at any node other than the root or the tip; whereas if $T_{\mathfrak{D}}$ forks at the node w and if v is one of the nodes immediately above w , then we may let T_1 consist of all the nodes below w , w itself, and all of the nodes above v and let T_2 consist of w and all of the remaining nodes not in T_1 .

Let $\mathfrak{E}$ and $\mathfrak{F}$ be the restrictions of the diagram $\mathfrak{D}$ to the T_1 and T_2 ; and let us assume that the anomalous node w_1 of $\mathfrak{D}$ belongs to $\mathfrak{F}$ (the proof for the case in which it belongs to $\mathfrak{E}$ is similar, but with the roles of $\mathfrak{E}$ and $\mathfrak{F}$ reversed). We now define the diagrams $\mathfrak{E}_n$ and $\mathfrak{F}_n$ by induction on $n = 0, 1, 2, \dots$:

$n = 0$. Set $\mathfrak{E}_0 = \mathfrak{E}$ and $\mathfrak{F}_0 = \mathfrak{F}$.

$n = 2k + 1, k \geq 0$. We let $\mathfrak{E}_{2k+1} = \mathfrak{E}_{2k}$ and let $\mathfrak{F}_{2k+1}$ be a regular extension of $\mathfrak{F}_{2k}$ whose new constants and s-predicate symbols do not belong to $\mathfrak{E}_{2k}$ and which contains no new original r-predicate symbols (if such an extension does not exist, we let $\mathfrak{F}_{2k+1}$ be $\mathfrak{F}_{2k}$).

$n = 2k, k > 0$. We let $\mathfrak{F}_{2k} = \mathfrak{F}_{2k-1}$ and let $\mathfrak{E}_{2k}$ be a regular extension of $\mathfrak{E}_{2k-1}$ whose new constants and s-predicate symbols do not belong to $\mathfrak{F}_{2k-1}$ and which contains no new original r-predicate symbols (if such an extension does not exist, we let $\mathfrak{E}_{2k}$ be $\mathfrak{E}_{2k-1}$).

Let $\mathfrak{D}_n = \mathfrak{E}_n \cup \mathfrak{F}_n$ for $n = 0, 1, 2, \dots$. Then we may show by a subsidiary induction on n that $\mathfrak{D}_n$ is an almost regular diagram whose sole anomalous node when $n > 0$ is w_0 and that $\mathfrak{F}_n$ is regular for odd n and $\mathfrak{E}_n$ regular for positive and even n :

$n = 0$. Trivial.

$n = 2k + 1, k \geq 0$. By IH (for n), $\mathfrak{D}_{2k}$ is an almost regular diagram whose sole anomalous node belongs to T_2 . Since $\mathfrak{F}_{2k}$ has fewer than m nodes, it follows by the main IH (for m) that $\mathfrak{F}_{2k}$ has a normal regular extension $\mathfrak{F}_{2k}^+$. Clearly, we may arrange for all of the new constants and s-predicate symbols of $\mathfrak{F}_{2k}^+$ not to occur in $\mathfrak{E}_{2k}$. So by the construction, $\mathfrak{F}_{2k+1}$ will be such an extension $\mathfrak{F}_{2k}^+$. But then, by Lemma 1, $\mathfrak{D}_{2k+1}$ will be an almost regular diagram whose sole anomalous node is w_0 .

$n = 2k, k > 1$. The proof is the same, mutatis mutandis, as for the previous case.

Let $\mathfrak{E}^+ = \mathfrak{E}_0 \cup \mathfrak{E}_1 \cup \mathfrak{E}_2 \cup \dots$, $\mathfrak{F}^+ = \mathfrak{F}_0 \cup \mathfrak{F}_1 \cup \mathfrak{F}_2 \cup \dots$, and $\mathfrak{D}^+ = \mathfrak{E}^+ \cup \mathfrak{F}^+$. Since each $\mathfrak{D}_n$, for $n > 0$ is a safe and almost regular extension of $\mathfrak{D}$ whose sole anomalous is w_0 , the same is true of $\mathfrak{D}^+$. But $\mathfrak{D}^+$ is also regular. For $\mathfrak{E}^+$ is the union of an increasing chain $\mathfrak{E}_2 \subseteq \mathfrak{E}_4 \subseteq \mathfrak{E}_6 \subseteq \dots$ of regular diagrams and so must itself be regular. Similarly for $\mathfrak{F}^+$. But then every arrow in $\mathfrak{D}^+ = \mathfrak{E}^+ \cup \mathfrak{F}^+$ is regular and so $\mathfrak{D}^+$ is itself regular. $\square$

We must now show how all possibilities may be ‘witnessed’ in a diagram. Say that a diagram $\mathfrak{D}$ is *M-complete* (‘M’ for possibility) if $\Diamond_{cF}, |B| \subset cF$, $cF \approx P \in \mathfrak{D}_w$ implies that, for some $v \in W_{\mathfrak{D}}$, $\langle w, v, P \rangle \in \mathfrak{D}$ and $B \in \mathfrak{D}_v$. Note that no requirement is directly imposed on $\Diamond_p B$ since it will not, in general, be a restricted sentence. A diagram is said to be *nice* if it is regular and M-complete.

THEOREM 3. *Any regular finite tree-diagram $\mathfrak{D}$ has a nice extension $\mathfrak{D}^+$.*

Proof. Let $w_0, w_1, w_2, \dots, a_0, a_1, a_2, \dots$ and $Q_0, Q_1, Q_2, \dots$ be respective enumerations of infinitely many world-constants, individual constants and s-predicate symbols that do not appear in $\mathfrak{D}$. Let $\mathbf{W} = \{w_0, w_1, w_2, \dots\}$, $\mathbf{A} = \{a_0, a_1, a_2, \dots\}$ and $\mathbf{Q} = \{Q_0, Q_1, Q_2, \dots\}$ and, for $\mathcal{L} = \mathcal{L}(\mathfrak{D})$, let $\mathcal{L}^+ = \mathcal{L} \cup \mathbf{A} \cup \mathbf{Q}$. Partition $\mathbf{A}$ and $\mathbf{Q}$ into infinitely many infinite sets $\mathbf{A}_0, \mathbf{A}_1, \mathbf{A}_2, \dots$ and $\mathbf{Q}_0, \mathbf{Q}_1, \mathbf{Q}_2, \dots$; and let $\mathcal{L}_n = \mathcal{L} \cup \mathbf{A}_0 \cup \mathbf{A}_1 \cup \dots \cup \mathbf{A}_n \cup \mathbf{Q}_0 \cup \mathbf{Q}_1 \cup \dots \cup \mathbf{Q}_n$ for $n = 0, 1, 2, \dots$. Let $M_0, M_1, M_2, \dots$ be a repeating enumeration of all quadruples of the form $\langle w, B, F, P \rangle$, where w is a world constant from $\mathbf{W}_{\mathfrak{D}} \cup \mathbf{W}$, B is an original sentence from $\mathcal{L}^+$, F is an original closed predicate from $\mathcal{L}^+$, and P is an s-predicate symbol from $\mathcal{L}^+$.

We define the diagrams $\mathfrak{D}_0, \mathfrak{D}_1, \mathfrak{D}_2, \dots$ as follows:

$n = 0$. $\mathfrak{D}_0 = \mathfrak{D}$.

$n = k + 1$. Suppose that $M_k = \langle w, B, F, P \rangle$. If it is not the case that $\Diamond_{cF}A, |B| \subset cF, cF \approx P \in \mathfrak{D}_{k,w}$, then let $\mathfrak{D}_{k+1} = \mathfrak{D}_k$. If it is the case that $\Diamond_{cF}B, |B| \subset cF, cF \approx P \in \mathfrak{D}_{k,w}$, then let $\mathfrak{D}_k^+ = \mathfrak{D}_k \cup \{\langle w, v, P \rangle, \langle v, B \rangle\}$, for v the first world-constant that does not appear in $\mathfrak{D}_k$, and let $\mathfrak{D}_{k+1}$ be a normal regular extension of $\mathfrak{D}_k^+$ in the language $\mathcal{L}_{k+1}$ (if no such extension exists, we may let $\mathfrak{D}_{k+1} = \mathfrak{D}_k^+$).

We may show by induction on $n = 0, 1, 2, \dots$ that $\mathfrak{D}_k$ is a regular diagram:

$n = 0$. Trivial.

$n = k + 1$. Let $\Delta = \mathfrak{D}_{k,w}$ and $\Gamma = \{B\}$. By Lemma VII.4, Γ is a P-companion to Δ . Now it is clear that the conditions for the application of Lemma VII.5 are satisfied; and so $\Gamma^+ = \Gamma \cup \{B : \Delta \vdash \Box_p B, B \text{ in } \mathcal{L}(\Delta)\}$ and Δ are matching P-companions. But then by Lemma VII.3, there are safe extensions Δ^* and Γ^* of Δ and Γ that are maximally matching P-companions and fully complete. Clearly, we can arrange that none of the new symbols of Δ^* or Γ^* occur in $\mathfrak{D}_k$. Let $\mathfrak{D}_k^+ = \mathfrak{D}_k \cup \{\langle w, C \rangle : C \in \Delta^*\} \cup \{C : \langle v, C \rangle \in \Gamma^*\} \cup \{\langle w, v, P \rangle\}$, for v a new world-constant. Then given by IH that $\mathfrak{D}_k$ is regular, it is readily verified that $\mathfrak{D}_k^+$ is almost regular with w the sole anomalous node. So by Lemma 2, it has a regular extension containing $\{\langle w, v, P \rangle, \langle v, B \rangle\}$; and $\mathfrak{D}_{k+1}$ is therefore such a regular extension.

Let $\mathfrak{D}^+ = \mathfrak{D}_0 \cup \mathfrak{D}_1 \cup \mathfrak{D}_2 \cup \dots$. Then since $\mathfrak{D}^+$ is the union of an increasing chain $\mathfrak{D}_0 \subseteq \mathfrak{D}_1 \subseteq \mathfrak{D}_2 \subseteq \dots$ of regular diagrams, it is itself a regular diagram. Suppose now that $\Diamond_{cF}B, |B| \subset cF, cF \approx P \in \mathfrak{D}_w$ for some world-constant w of $\mathfrak{D}^+$. Then, for some k , $\Diamond_{cF}B, |B| \subset cF$,

$cF \approx P \in \mathfrak{D}_{k,w}$. But, for some $l > k$, $M_l = \langle w, B, F, P \rangle$. So by the construction, there is a v for which $\langle w, v, P \rangle, \langle v, B \rangle \in \mathfrak{D}_{k+1} \subseteq \mathfrak{D}^+$. $\square$

COROLLARY 4. *Suppose that Δ is a consistent set of sentences with original individual constants and predicate symbols. There is then a nice diagram with root w_0 for which $\mathfrak{D}_{w_0}$ contains Δ .*

Proof. By Theorem 3 and Lemma VII.6. $\square$

9. THE CANONICAL MODEL

We construct a canonical model from a nice diagram in the obvious way and show that it verifies at each world the very sentences belonging to that world. We first give a breakdown of the results we shall require.

The truth of $\Box$ -sentences is transferred along appropriate arrows:

LEMMA 1. *Suppose that $\mathfrak{D}$ is a nice diagram and that $\langle w, v, P \rangle \in \mathfrak{D}$. Then:*

- (i) $\mathfrak{D}_w \vdash \Box_p B$ with B in $\mathcal{L}(\mathfrak{D}_w)$ implies $B \in \mathfrak{D}_v$;
- (ii) $\Box_p B \in \mathfrak{D}_w$ implies $\Box_p B \in \mathfrak{D}_v$.

Proof. (i) Suppose $\mathfrak{D}_w \vdash \Box_p B$ with B in $\mathcal{L}(\mathfrak{D}_w)$. Since $\mathfrak{D}_w$ and $\mathfrak{D}_v$ exactly P-match, B is in $\mathcal{L}(\mathfrak{D}_v)$. But then given that $\mathfrak{D}_v$ is a maximal P-companion to $\mathfrak{D}_w$, $B \in \mathfrak{D}_v$.

(ii) Suppose $\Box_p B \in \mathfrak{D}_w$. Then $\Box_p \Box_p B$ is in $\mathcal{L}(\mathfrak{D}_w)$ (though not necessarily restricted) and, by Theorem IV.6(ii) of LE, $\mathfrak{D}_w \vdash \Box_p \Box_p B$. So given that $\Box_p B$ is among the restricted sentences of $\mathcal{L}(\mathfrak{D}_w)$, it follows by (i) that $\Box_p B \in \mathfrak{D}_v$. $\square$

The r-predicate symbols behave as rigid predicates:

LEMMA 2 (Predicate Rigidity). *Let $\mathfrak{D}$ be a nice diagram and P an r-predicate symbol. Then:*

- (i) if $Qa \in \mathfrak{D}_w$ and $Q \in \mathcal{L}(\mathfrak{D}_v)$ then $Qa \in \mathfrak{D}_v$;
- (ii) if $Q \in \mathcal{L}(\mathfrak{D}_w)$ and $a \in \mathcal{L}(\mathfrak{D}_v)$ whenever $Qa \in \mathfrak{D}_w$ then $Q \in \mathcal{L}(\mathfrak{D}_v)$.

Proof. Let $w = w_0, w_1, \dots, w_n = v$ be the path from w to v in $T_{\mathfrak{D}}$. The result, in both cases, is proved by induction on n :

(i) $n = 0$. Trivial.

$n = k + 1$. Suppose $Qa \in \mathfrak{D}_w$ and $Q \in \mathcal{L}(\mathfrak{D}_v)$. Since $Q \in \mathcal{L}(\mathfrak{D}_w) \cap \mathcal{L}(\mathfrak{D}_v)$, it follows by the path condition that $Q \in \mathcal{L}(\mathfrak{D}_{w_k})$. So by

IH, $Qa \in \mathfrak{D}_{w_k}$. Suppose w_k is linked by P to $w_{k+1} = v$ in $\mathfrak{D}$. Then since $Q \in \mathcal{L}(\mathfrak{D}_{w_k}) \cap \mathcal{L}(\mathfrak{D}_{w_{k+1}})$, Q is P -bound in $\mathfrak{D}_{w_k}$, i.e. $Q \subset P \in \mathfrak{D}_{w_k}$. So from $Qa \in \mathfrak{D}_{w_k}$ it follows by subsumption that $\mathfrak{D}_{w_k} \vdash \Box_P Qa$. But then $Qa \in \mathfrak{D}_{w_{k+1}}$ by Lemma 1.

(ii) $n = 0$. Trivial.

$n = k + 1$. Suppose $Q \in \mathcal{L}(\mathfrak{D}_w)$ and $a \in \mathcal{L}(\mathfrak{D}_v)$ whenever $Qa \in \mathfrak{D}_w$. Then by the path condition, $a \in \mathcal{L}(\mathfrak{D}_{w_k})$ whenever $Qa \in \mathfrak{D}_w$. So $Q \in \mathcal{L}(\mathfrak{D}_{w_k})$ by IH and hence, by part (i), $Qa \in \mathfrak{D}_{w_k}$ iff $Qa \in \mathfrak{D}_w$. Suppose that w_k is linked by P to w_{k+1} in $\mathfrak{D}$. Then $Q \subset P \in \mathfrak{D}_{w_k}$. For otherwise, by $\mathfrak{D}_{w_k}$ N -complete, $\sim(Q \subset P) \in \mathfrak{D}_{w_k}$; and so, by $\mathfrak{D}_{w_k}$ Q -complete, $Qa, \sim Pa \in \mathfrak{D}_{w_k}$ for some constant a . Since $Qa \in \mathfrak{D}_{w_k}$, $Qa \in \mathfrak{D}_w$ and hence, by supposition, $a \in \mathcal{L}(\mathfrak{D}_v)$. But then since $a \in \mathcal{L}(\mathfrak{D}_{w_k}) \cap \mathcal{L}(\mathfrak{D}_{w_{k+1}})$, it follows from the P -match between $\mathfrak{D}_{w_k}$ and $\mathfrak{D}_{w_{k+1}}$ that a is P -bound in $\mathfrak{D}_{w_k}$, contrary to the fact that $\sim Pa \in \mathfrak{D}_{w_k}$. Therefore $Q \subset P \in \mathfrak{D}_{w_k}$ and so, again, by the P -match between $\mathfrak{D}_{w_k}$ and $\mathfrak{D}_{w_{k+1}}$, $Q \in \mathcal{L}(\mathfrak{D}_{w_{k+1}})$.

Identity and dependence behave as they should:

LEMMA 3. *Let $\mathfrak{D}$ be a nice diagram. Then:*

- (i) $a = b \in \mathfrak{D}_w$ and $a \in \mathcal{L}(\mathfrak{D}_v)$ implies $a = b \in \mathfrak{D}_v$;
- (ii) $a \geq b \in \mathfrak{D}_w$ and $a \in \mathcal{L}(\mathfrak{D}_v)$ implies $a \geq b \in \mathfrak{D}_v$;
- (iii) $B(a) \in \mathfrak{D}_w$ and $a = b \in \mathfrak{D}_v$ implies $B(b) \in \mathfrak{D}_w$.

Proof. (i) Let $w = w_0, w_1, \dots, w_n = v$ be the path from w to v in $T_{\mathfrak{D}}$. We prove that $a = b \in \mathfrak{D}_{w_n}$ by induction on n :

$n = 0$. Trivial.

$n = k + 1$. By the Path Condition, $a \in \mathcal{L}(\mathfrak{D}_{w_k})$. So by IH, $a = b \in \mathfrak{D}_{w_k}$. Suppose that w_k is linked to w_{k+1} by P in $\mathfrak{D}$. Since $a \in \mathcal{L}(\mathfrak{D}_{w_k}) \cap \mathcal{L}(\mathfrak{D}_{w_{k+1}})$, it follows from the P -match between $\mathfrak{D}_{w_k}$ and $\mathfrak{D}_{w_{k+1}}$ that $Pa \in \mathfrak{D}_{w_k} \cap \mathfrak{D}_{w_{k+1}}$. By Theorem IV.7(i) of LE, $\vdash a = b \rightarrow \Box_a a = b$; so by subsumption, $Pa, a = b \vdash \Box_a a = b$; and so, $\Box_a a = b, Pb \in \mathfrak{D}_{w_k}$. But then $a = b \in \mathfrak{D}_{w_{k+1}}$ by Lemma 1.

(ii) Again, let $w = w_0, w_1, \dots, w_n = v$ be the path from w to v in $T_{\mathfrak{D}}$. We prove that $a \geq b \in \mathfrak{D}_{w_n}$ by induction on m .

$m = 0$. Trivial.

$m = k + 1$. By the Path Condition, $a \in \mathcal{L}(\mathfrak{D}_{w_k})$. So by IH, $a \geq b \in \mathfrak{D}_{w_k}$. Suppose that w_k is linked to w_{k+1} by P in $\mathfrak{D}$. Since $a \in \mathcal{L}(\mathfrak{D}_{w_k}) \cap \mathcal{L}(\mathfrak{D}_{w_{k+1}})$, it follows from the P -match between $\mathfrak{D}_{w_k}$ and $\mathfrak{D}_{w_{k+1}}$ that $Pa \in \mathfrak{D}_{w_k} \cap \mathfrak{D}_{w_{k+1}}$. By Theorem IV.10(iii) of LE, $\vdash a \geq b \rightarrow \Box_a a \geq b$; so by Subsumption and Chaining, $Pa, a \geq b \vdash \Box_a a \geq b$; and so, $\Box_a a \geq b \in \mathfrak{D}_{w_k}$. But then $a \geq b \in \mathfrak{D}_{w_{k+1}}$ by Lemma 1.

(iii) Suppose $B(a) \in \mathcal{D}_w$ and $a = b \in \mathcal{D}_v$. If a does not occur in $B(a)$, there is nothing to prove. If it does, then $a \in \mathcal{L}(\mathcal{D}_w)$. But then $a = b \in \mathcal{D}_w$ by part (i); and so $B(b) \in \mathcal{D}_w$ by the logic of identity. $\square$

It is now easy to show that '=' behaves in the manner of an equivalence relation on constants. Let $K_{\mathcal{D}}$ be the set of constants that appear in $\mathcal{D}$. Define the relation $\equiv$ on $K_{\mathcal{D}}$ by:

$$a \equiv b \text{ if } a = b \in \mathcal{D}_w \text{ for some } w \text{ in } W_{\mathcal{D}}.$$

LEMMA 4. $\equiv$ is an equivalence relation.

Proof. The verification of reflexivity and symmetry is straightforward. To establish transitivity, suppose that $a = b \in \mathcal{D}_w$ and $b = c \in \mathcal{D}_v$. Then $b = a \in \mathcal{D}_w$ by the logic of identity. Since $b \in \mathcal{L}(\mathcal{D}_w)$, $b = c \in \mathcal{D}_w$ by Lemma 3(i); and so, again by the logic of identity, $a = c \in \mathcal{D}_w$. $\square$

We can now define the canonical model for a nice diagram $\mathcal{D}$. Given $a \in K_{\mathcal{D}}$, let $a^* = \{b \in K_{\mathcal{D}} : b \equiv a\}$. The canonical model $M_{\mathcal{D}}$ for $\mathcal{D}$ is then the quadruple $\langle W, I, \succeq, \phi \rangle$, where:

- (i) $W = W_{\mathcal{D}}$;
- (ii) $I_w = \{a^* : a \in \mathcal{L}(\mathcal{D}_w)\}$;
- (iii) $\succeq = \{\langle a^*, b^* \rangle : a \geq b \in \mathcal{D}_w \text{ for some } w \in W\}$;
- (iv) $\phi(a) = a^*$ for each constant,
 $\phi(w, F) = \{\langle a_1^*, \dots, a_n^* \rangle : Fa_1 \dots a_n \in \mathcal{D}_w\}$ for each pure n -place predicate symbol F , and
 $\phi(P) = \{a^* : Pa \in \mathcal{D}_w\}$ for some $w \in W\}$ for each rigid predicate symbol P of $\mathcal{D}$.

Note that the canonical model is full; each object has a name.

LEMMA 5. For $\mathcal{D}$ nice, $M_{\mathcal{D}}$ is indeed a model.

Proof. $\Sigma x(x = x) \in \mathcal{D}_w$ for any w . By $\mathcal{D}_w$ Q-complete, $a = a \in \mathcal{D}_w$; and so each $I_w = \{a^* : a \in \mathcal{L}(\mathcal{D}_w)\}$ is non-empty. It remains to show that each I_w is closed under $\succeq$. So suppose $a^* \in I_w$ and $a^* \succeq b^*$. Then $a' \geq b' \in \mathcal{D}_v$ for some $v \in W_{\mathcal{D}}$, where $a' \equiv a$ and $b' \equiv b$. So $a' \in \mathcal{L}(\mathcal{D}_w)$ by Lemma 3(i) and $a' \geq b' \in \mathcal{D}_w$ by Lemma 3(ii). But then $b' \in \mathcal{L}(\mathcal{D}_w)$ and $b^* \in I_w$, as required. $\square$

Sentences and predicates are defined at a world in the canonical model when they are in the language of that world:

LEMMA 6 (Definability). Let $\mathcal{D}$ be a nice diagram. Then a sentence or closed predicate E is defined at w in the associated canonical $M_{\mathcal{D}}$ iff E is in $\mathcal{L}(\mathcal{D}_w)$.

Proof. E is defined at $w \in W_{\mathcal{D}}$ iff $\phi(a) \in I_w$ and $\phi(P) \subseteq I_w$ for each constant a and r -predicate symbol P in E . E is in $\mathcal{L}(\mathcal{D}_w)$ iff $a \in \mathcal{L}(\mathcal{D}_w)$ and $P \in \mathcal{L}(\mathcal{D}_w)$ for each constant a and r -predicate symbol P in E . Therefore it suffices to show:

- (i) $\phi(a) \in I_w$ iff $a \in \mathcal{L}(\mathcal{D}_w)$; and
- (ii) $\phi(P) \subseteq I_w$ iff $P \in \mathcal{L}(\mathcal{D}_w)$.

We consider each in turn:

- (i) $\phi(a) \in I_w$ iff $a = b \in \mathcal{D}_v$ for some v and some $b \in \mathcal{L}(\mathcal{D}_w)$ iff $a \in \mathcal{L}(\mathcal{D}_w)$ by Lemma 3(i).
- (ii) Suppose that $P \in \mathcal{L}(\mathcal{D}_v)$ for $v \in W_{\mathcal{D}}$. Then:

$$\begin{aligned}\phi(P) &= \{a^* : Pa \in \mathcal{D}_u \text{ for some } u \in W_{\mathcal{D}}\} \\ &= \{a^* : Pa \in \mathcal{D}_v\} \quad \text{by Lemma 2(i).}\end{aligned}$$

So:

$$\begin{aligned}\phi(P) \subseteq I_w &\quad \text{iff } \{a^* : Pa \in \mathcal{D}_v\} \subseteq I_w \\ &\quad \text{iff } \{a : Pa \in \mathcal{D}_v\} \subseteq \mathcal{L}(\mathcal{D}_w) \text{ by Lemma 3(i)} \\ &\quad \text{iff } P \in \mathcal{L}(\mathcal{D}_w) \text{ by Lemma 2(ii).} \quad \square\end{aligned}$$

THEOREM 7 (Truth). *Let $\mathcal{D}$ be a nice diagram and $M_{\mathcal{D}} = \langle W, I, \succeq, \phi \rangle$ its canonical model. Then for each $w \in W_{\mathcal{D}}$, sentence B and closed n -place predicate H :*

- (a) $w \models B$ if $B \in \mathcal{D}_w$, for any B in $\mathcal{L}(\mathcal{D}_w)$; and
- (b) $H_w = \{ \langle a_1^*, \dots, a_n^* \rangle : Ha_1 \dots a_n \in \mathcal{D}_w \}$, for any H in $\mathcal{L}(\mathcal{D}_w)$.

Proof. We should note that by Lemma 6 the stipulated side conditions under (a) and (b) are exactly those in which the sentence B and predicate H will be defined at w in $M_{\mathcal{D}}$.

Normally, for the purposes of a proof by induction, we would replace (a) by the corresponding biconditional:

$$(a)^+ \quad w \models B \text{ iff } B \in \mathcal{D}_w, \text{ for any } B \text{ in } \mathcal{L}(\mathcal{D}_w).$$

But this is only because B may figure in other more complex sentences. In the present case, unnegatable sentences of the form $\Box_s C$ do not occur in more complex sentences; and so it suffices to establish $(a)^+$ whenever B is negatable and (a) whenever B is unnegatable.

The proof now proceeds by a simultaneous induction on B and G :

(i) (a) $B = Ha_1 \dots a_n$.

$$\begin{aligned} w \models Ha_1 \dots a_n & \quad \text{iff } \langle \phi(a_1), \dots, \phi(a_n) \rangle \in H_w \\ & \quad \text{iff } \langle a_1^*, \dots, a_n^* \rangle \in H_w, \text{ by def. of } \phi \\ & \quad \text{iff } Ha_1 \dots a_n \in \mathcal{D}_w \text{ by IH (b).} \end{aligned}$$

(i) (b) $B = (a = b)$.

$$\begin{aligned} w \models a = b & \quad \text{iff } \phi(a) = \phi(b) \text{ iff} \\ & \quad \text{iff } a^* = b^* \\ & \quad \text{iff } a = b \in \mathcal{D}_v \text{ for some } v \in W_{\mathcal{D}} \\ & \quad \text{iff } a = b \in \mathcal{D}_w \text{ by Lemma 3(i) and the fact that} \\ & \quad a = b \text{ is in } \mathcal{L}(\mathcal{D}_w). \end{aligned}$$

(i) (c)

$$\begin{aligned} w \models a \geq b & \quad \text{iff } \phi(a) \succeq \phi(b) \\ & \quad \text{iff } a^* \succeq b^* \\ & \quad \text{iff } a \geq b \in \mathcal{D}_v \text{ for some } v \in W_{\mathcal{D}} \\ & \quad \text{iff } a \geq b \in \mathcal{D}_w \text{ by Lemma 3(ii) and the fact that} \\ & \quad a \geq b \text{ is in } \mathcal{L}(\mathcal{D}_w). \end{aligned}$$

(ii) $B = \sim C$. By $\mathcal{D}_w$ consistent and N-complete;

(iii) $B = (C \vee D)$. Again, by $\mathcal{D}_w$ consistent and N-complete;

(iv) $B = \prod x C(x)$. By $\mathcal{D}_w$ standardly complete.

(v) $B = \Box_F C$. Suppose first that $\Box_F C \in \mathcal{D}_w$. Since $\mathcal{D}_w$ is P-complete, $F \approx Q \in \mathcal{D}_w$ for some s-predicate symbol Q (if F is already an s-predicate symbol, we may let $Q = F$). Let $J = F_w$. By IH, $J = \{a^* : Fa \in \mathcal{D}_w\}$; and, given $F \approx Q \in \mathcal{D}_w$, it readily follows that $J = \{a^* : Qa \in \mathcal{D}_w\}$. Now suppose $J \subseteq I_v$, i.e. $\{a^* : Qa \in \mathcal{D}_w\} \subseteq \{a^* : a \in \mathcal{L}(\mathcal{D}_v)\}$. Then by Lemmas 2(i) and 3(i), $\{a : Qa \in \mathcal{D}_w\} \subseteq \{a : a \in \mathcal{L}(\mathcal{D}_v)\}$; and so by Lemma 2(ii), $Q \in \mathcal{L}(\mathcal{D}_v)$.

Let $w = w_0, w_1, \dots, w_n = v$ be a path from w to v . We now prove by induction on that $\Box_Q C \in \mathcal{D}_{w_m}$, $m = 0, 1, \dots, n$:

$m = 0$. Trivial.

$m = k + 1$. By IH, $\Box_Q C \in \mathcal{D}_{w_k}$. By the path condition, $Q \in \mathcal{L}(\mathcal{D}_{w_{k+1}})$. Suppose that w_k is linked to w_{k+1} by P in $T_{\mathcal{D}}$. Since $Q \in \mathcal{L}(\mathcal{D}_{w_k}) \cap \mathcal{L}(\mathcal{D}_{w_{k+1}})$, $Q \subset P \in \mathcal{D}_{w_k}$. So by Theorem IV.6(ii) of LE and subsumption, $\mathcal{D}_{w_k} \vdash \Box_P \Box_Q C$; and so by Lemma 1(ii), $\Box_Q C \in \mathcal{D}_{w_{k+1}}$.

Letting $m = n$ in the above result, $\Box_Q C \in \mathcal{D}_v$ and so $C \in \mathcal{D}_v$. But then by IH, $v \models C$; and so $w \models \Box_F C$.

Suppose now that $\Box_F C \notin \mathfrak{D}_w$. Let us note that in this case F will be an original predicate and so, by $\mathfrak{D}_w$ an N-complete theory, $\Diamond_F \sim C \in \mathfrak{D}_w$. We distinguish two subcases.

(1) $|C| \subset cF \notin \mathfrak{D}_w$. By $\mathfrak{D}_w$ N-complete and the fact that C is an original sentence, $\sim(|C| \subset cF) \in \mathfrak{D}_w$. It now follows from IH (and an unproblematic application of (ii)–(iv)) that $w \models \sim(|C| \subset cF)$. But $\vdash \sim(|C| \subset cF) \rightarrow \Box_F C$; and so, by Soundness, not $w \models \Box_F C$.

(2) $|C| \subset cF \in \mathfrak{D}_w$. By axiom (IV)(i), $\Diamond_{cF} \sim C \in \mathfrak{D}_w$. By $\mathfrak{D}_w$ P-complete, $cF \approx Q \in \mathfrak{D}_w$ for some s-predicate symbol Q . So by $\mathfrak{D}$ M-complete, there is a v for which $\langle w, v, Q \rangle \in \mathfrak{D}$ and $\sim C \in \mathfrak{D}_v$. By $\mathfrak{D}_v$ consistent, $C \notin \mathfrak{D}_v$; and by IH, not $v \models C$.

Let $J = F_w$. Then

$$\begin{aligned} c(J) &= c(F_w) = (cF)_w \\ &= \{a^* : (cF)a \in \mathfrak{D}_w\}, \quad \text{from IH} \\ &= \{a^* : Qa \in \mathfrak{D}_w\}, \quad \text{by } cF \approx Q \in \mathfrak{D}_w. \end{aligned}$$

Since $Qa \in \mathfrak{D}_w$ implies $Qa \in \mathfrak{D}_v$, $c(J) \subseteq I_v$ and so $J \subseteq c(J) \subseteq I_v$. Therefore, given not $v \models C$, not $w \models \Box_F C$.

(vi) H a pure predicate symbol F .

Then

$$\begin{aligned} F_w &= \phi(w, F) \quad \text{by definition of } F_w \\ &= \{a^* : Fa \in \mathfrak{D}_w\} \quad \text{by definition of } \phi \text{ in } M_{\mathfrak{D}}. \end{aligned}$$

(vii) H a rigid predicate symbol P .

Then

$$\begin{aligned} P_w &= \phi(P) \quad \text{by definition of } P_w \\ &= \{a^* : Pa \in \mathfrak{D}_v \text{ for some } v \in W_{\mathfrak{D}}\} \\ &= \{a^* : Pa \in \mathfrak{D}_w\} \text{ by Lemma 2(i) and the fact that} \\ &\quad P \in \mathcal{L}(\mathfrak{D}_w). \end{aligned}$$

(viii) H a λ -abstract $\lambda x A(x)$.

$$\begin{aligned} \lambda x A(x)_w &= \{a^* \in I_w : w \models A(a)\} \quad \text{by definition of } \lambda x A(x)_w \\ &= \{a^* \in I_w : A(a) \in \mathfrak{D}_w\} \quad \text{by IH} \\ &= \{a^* \in I_w : \lambda x A(x)a \in \mathfrak{D}_w\} \quad \text{by Axiom III(i).} \end{aligned}$$

□

From Theorem 7 and Corollary VIII.4 we obtain our main result:

THEOREM 8. *Any consistent set of formulas of E5 has a model.*

This result can be extended to various stronger systems in an obvious way. Suppose, for example, that we add the domain axiom:

$$(V) (ii) \quad \prod xPx \rightarrow \Box_p \prod xPx$$

of LE. Then the resulting system can be proved complete for the condition that no domain of a model be properly included in any other domain. Or again, if an existence-predicate E is added to the language, then the axioms:

$$(V) (i) \quad \prod xPx \rightarrow \Diamond_p Ex; \quad \text{and}$$

$$(\geq E) \quad x \geq y \rightarrow (Ex \rightarrow Ey),$$

correspond to the conditions:

$$(V) (i)^* \quad \text{if } a \in I_w \text{ then, for some } v \in W, I_v \supseteq I_w \text{ and } a \in \phi(E, v);$$

$$(\geq E)^* \quad b \in \phi(w, E) \text{ whenever } a \in \phi(w, E) \text{ and } a \geq b.$$

It would be of interest to extend the completeness results to the variants of E5 considered at the end of LE and also to systems with a different underlying modality. For some of these variants, the proofs of completeness are relatively straightforward;² for others it is not. We should also note that there are other logics in which the modal operator may be naturally taken to be indexed to a group of items. We may talk, for example, of group knowledge or of group obligation or of what is provable by a group of proofs, i.e. within a system. It would be of interest to work out systems for these other cases and to develop a general account of systems of this sort.

NOTES

¹ I should like to thank the referee and participants of conferences at Stanford (94) and Uppsala (98) for helpful remarks on an earlier presentation of this material.

² Let me note, in this connection, that Fabrice Correia has established completeness for a propositional version of the system using a direct canonical model construction and Joshua Schechter has done the same for various systems with a weak underlying modality.

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